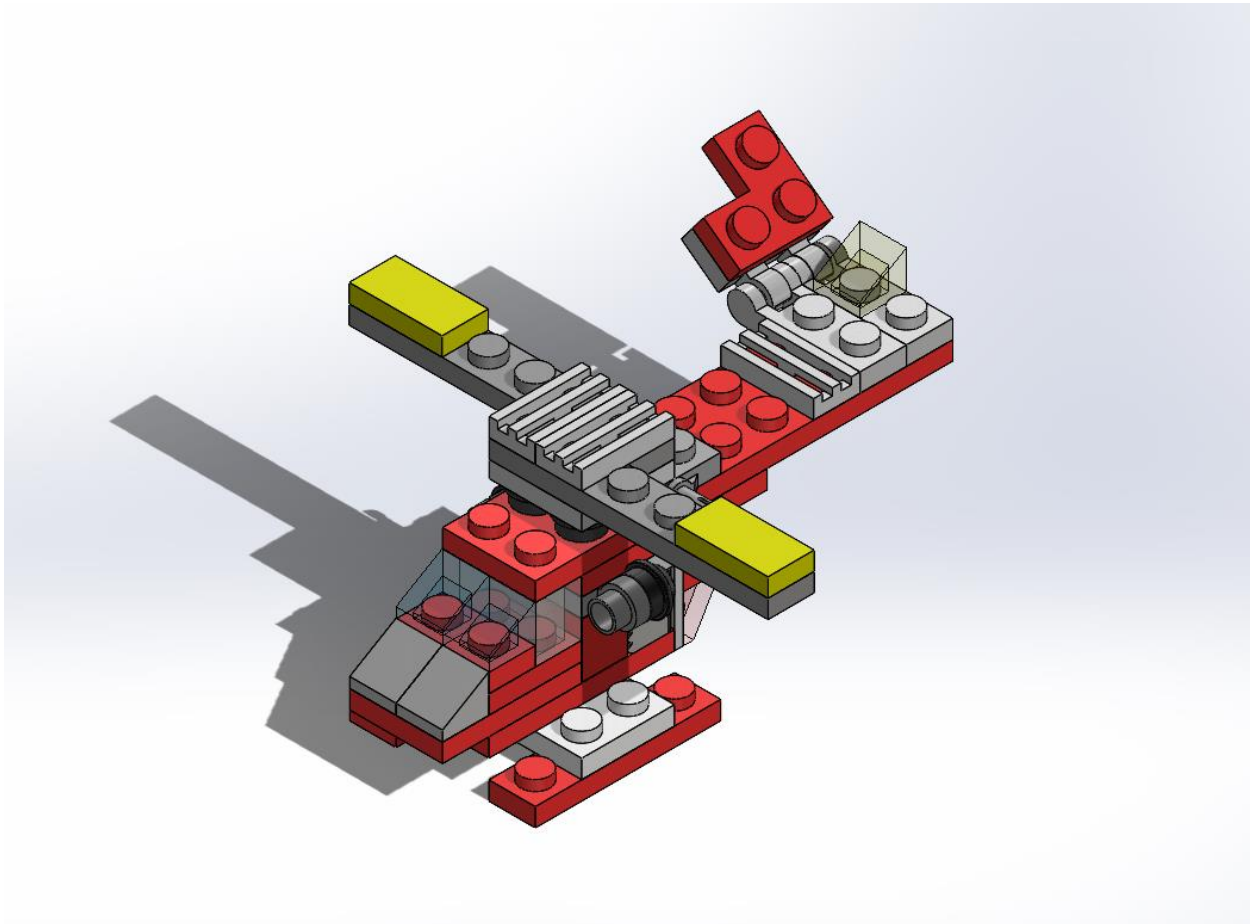


LEGO Project Helicopter
ME187 – Graphical Communications
Boise State University

Jennifer Sennett



Milestone # 1 – Reverse Engineering

Name: Jennifer Sennett

Date: January 19, 2024

Customer Problem Statement:

The customer, SafeOutdoors INC, needs a flight vehicle that is durable, has a flight capacity to reach locations within their city limits and on the outskirts of their city limits, is usable in the dark, has an enclosed cab for comfort and has safety features.

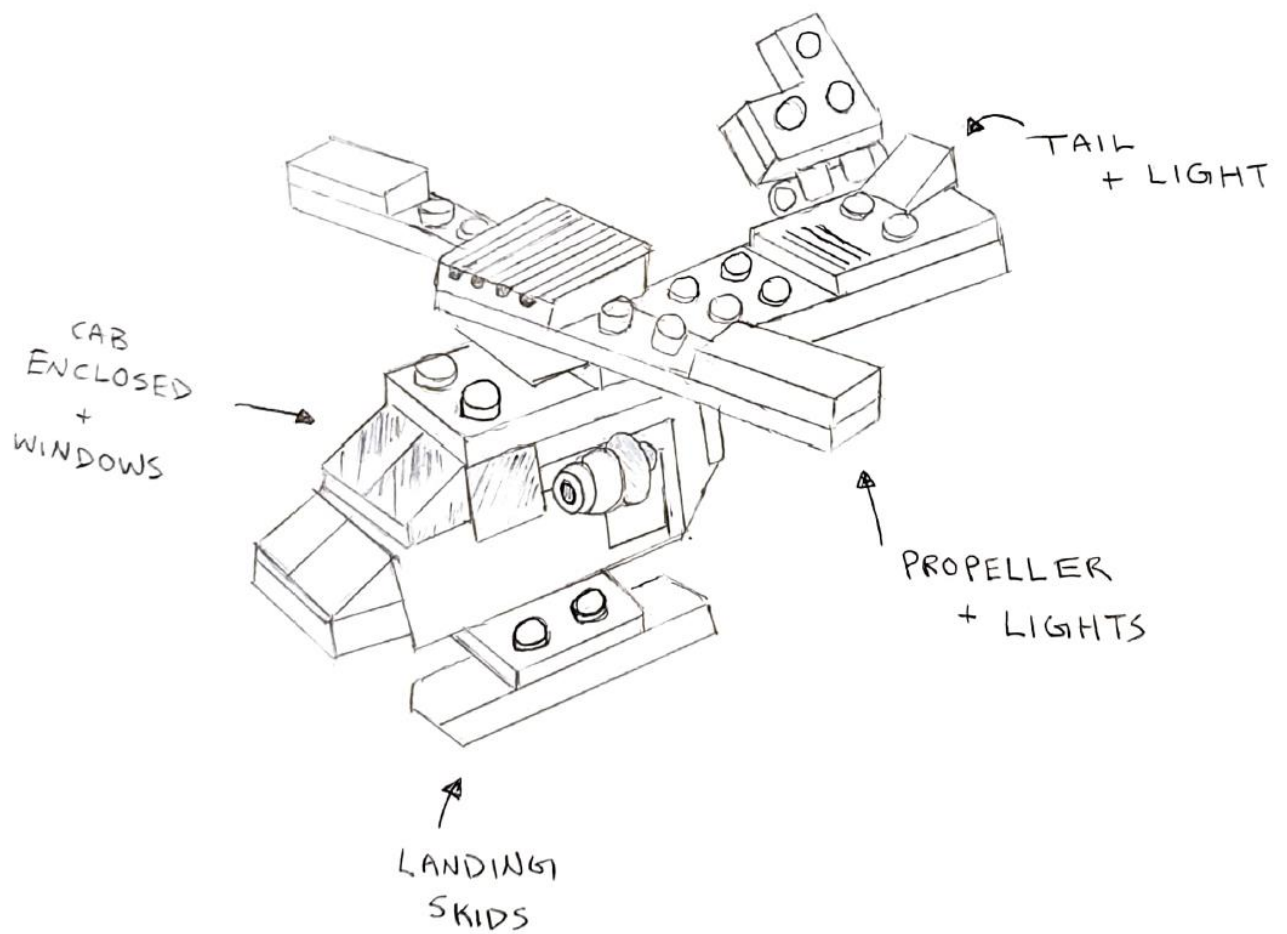
Project Objectives:

- Durable
- Travel city limits/outskirts of city limits
- Functions at night or when dark
- Enclosed cab for comfort
- Safety features

Preliminary Ideas:

Durability/Enclosed Cab	Night/Flight Range Accessories	Engine/Carrying-Capacity	Safety and Comfort Features
➤ Metallic micro lattice material interior cage/paneling ➤ Aluminum-alloy body/exterior paneling ➤ Impact/heat resistant plastic windows	➤ High-powered spotlight ➤ Flood lights ➤ Detachable 100 gallon exterior water tank ➤ Adjustable, carbon fiber propellers	➤ 5 seat cabin capacity ➤ 100 gallon on-board water or retardant tank ➤ 75 gallon, piston engine	➤ Steel frame ➤ Heat-resistant rubber skids ➤ Heat-resistant exterior coating ➤ State-of-the-art GPS navigation system and auto-pilot ➤ Air-conditioning

Sketch

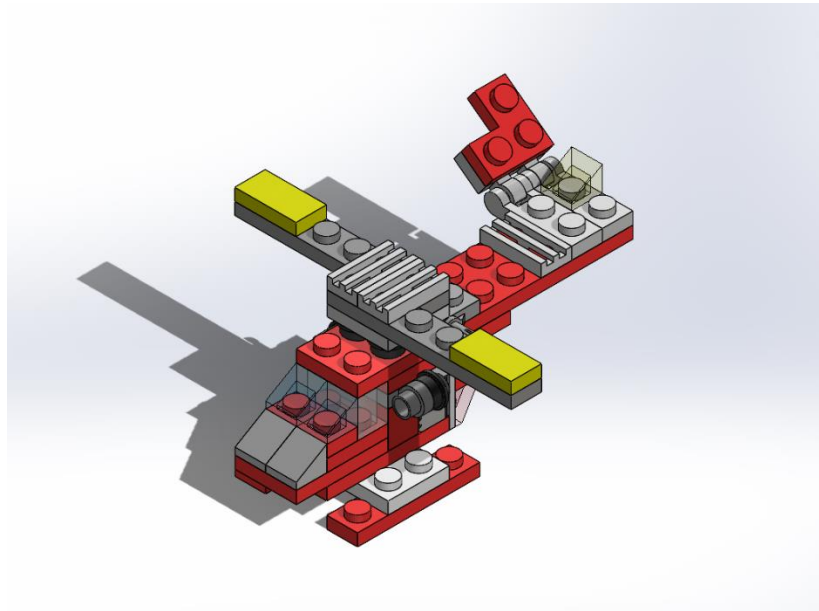


Summary

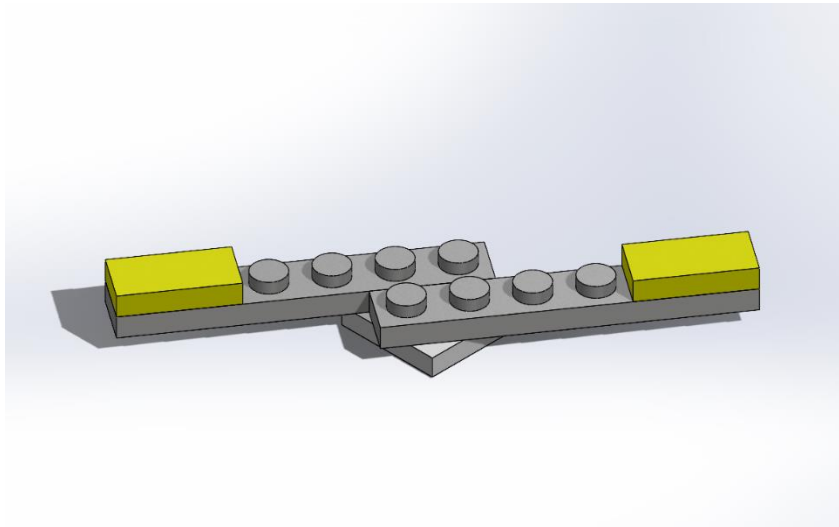
All LEGO pieces are specifically designed and molded so that each piece, regardless of how old, new, or if they are from different sets can fit together seamlessly. They all have the same size circular-snapping components with male and female ends, which allow them to snap together uniformly, regardless of their length, depth, or characteristics (such as having different edges, and shapes, color and transparency). These snapping components, when fit together, always form a grid. These features make designing the assembled object in CAD software easier and will also allow for simpler disassembly when breaking out each part individually in a technical drawing.

Signature: Jennifer Sennett

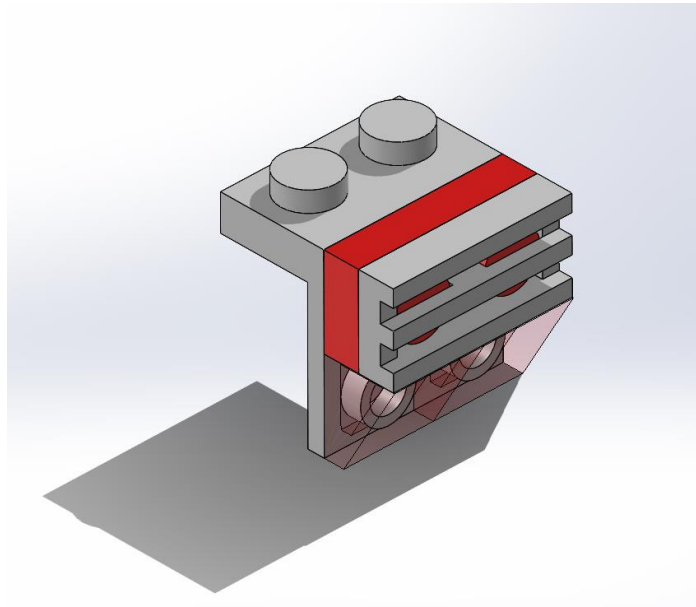
Final Assembly



Propeller Assembly



Cab Back Assembly

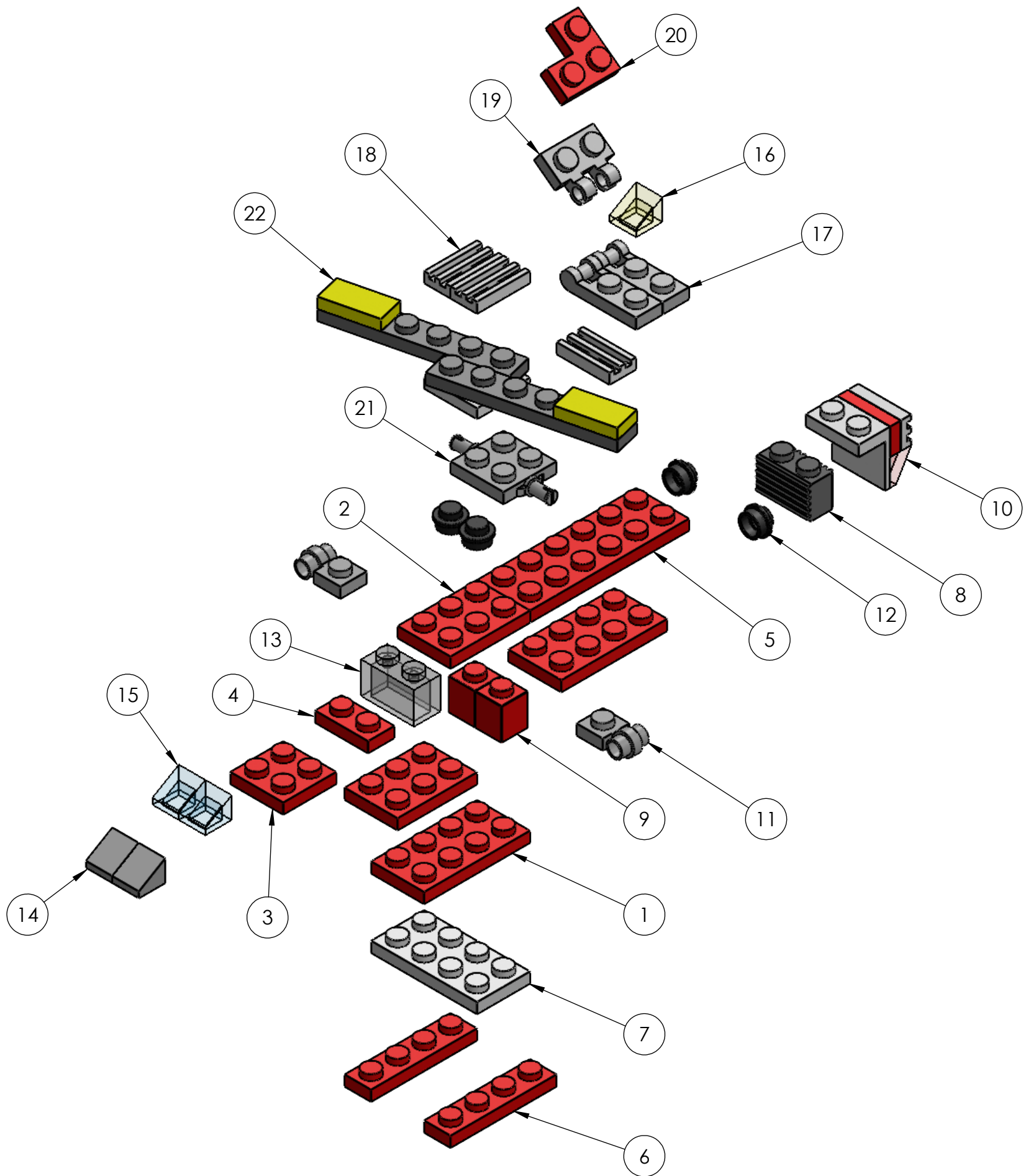


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4	302321	1
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6	371021	2
7	302001	1
8	4210636	1
9	300521	2
10	CabBack Assembly	1
11	4632575	2
12	614126	4
13	4226179	1
14	4521921	2
15	4244366	2
16	4260942	1
17	4515369	2
18	4211350	3
19	4556157	1
20	242021	1
21	4211504	1
22	Propeller Assembly	1

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MATERIAL: ABS PLASTIC		
WEIGHT:		

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SAFEOUTDOORS INC.		
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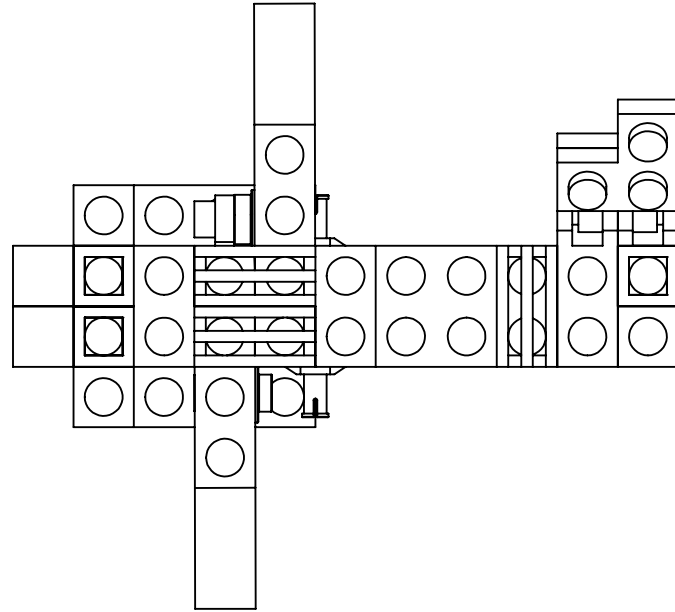
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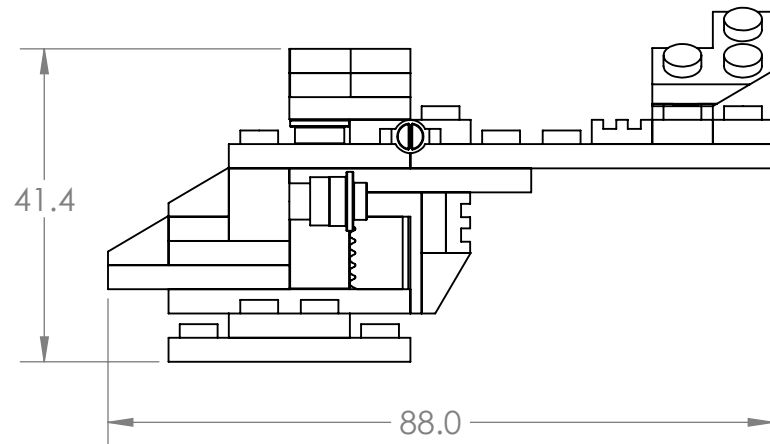
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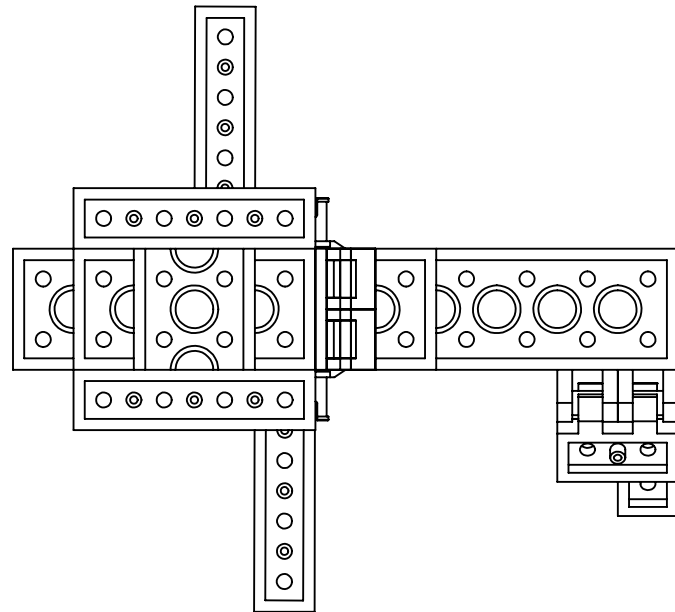
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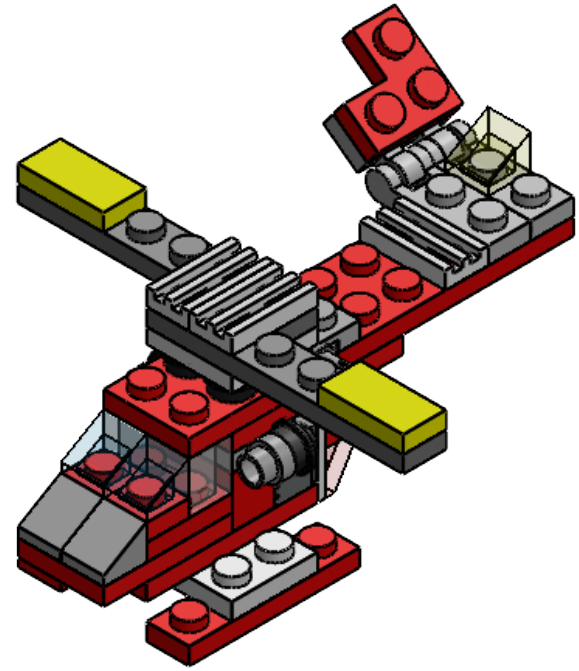
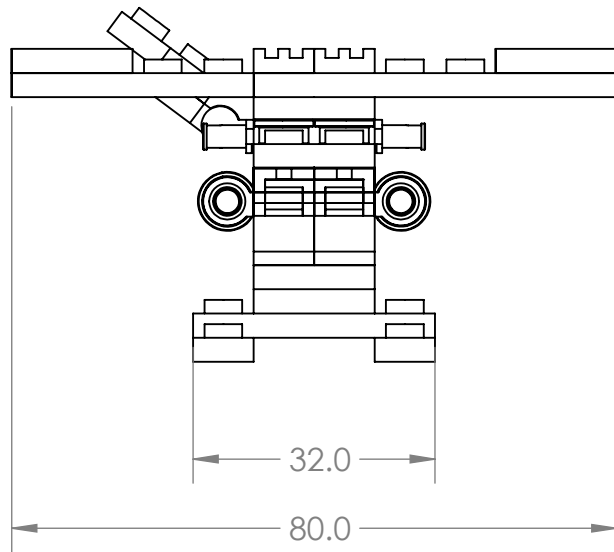
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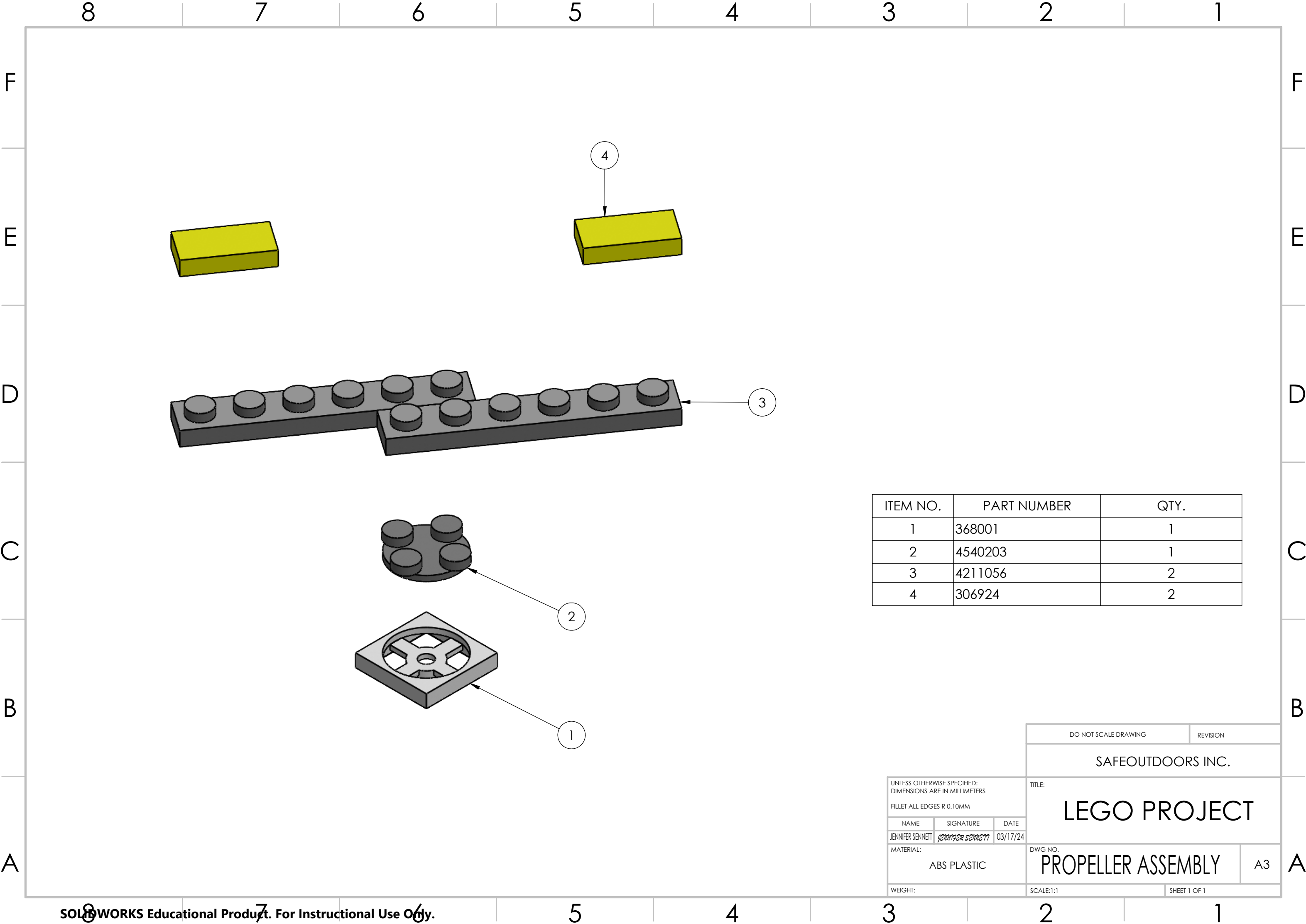


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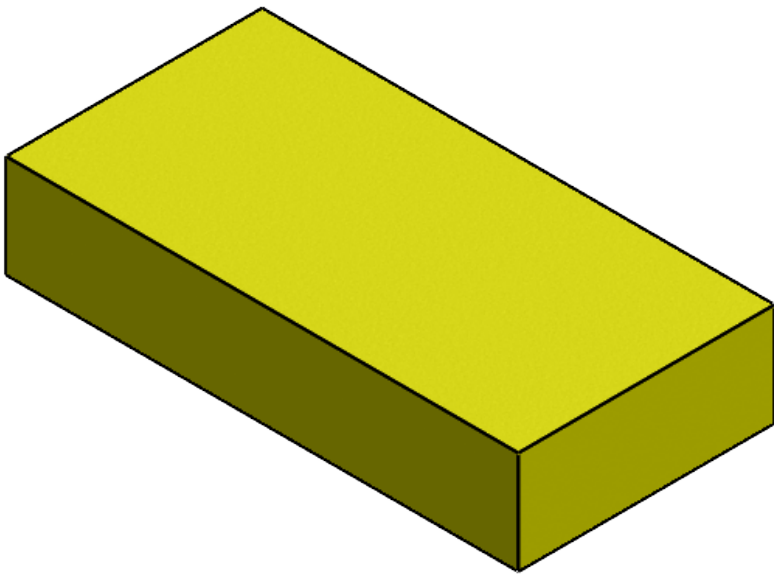
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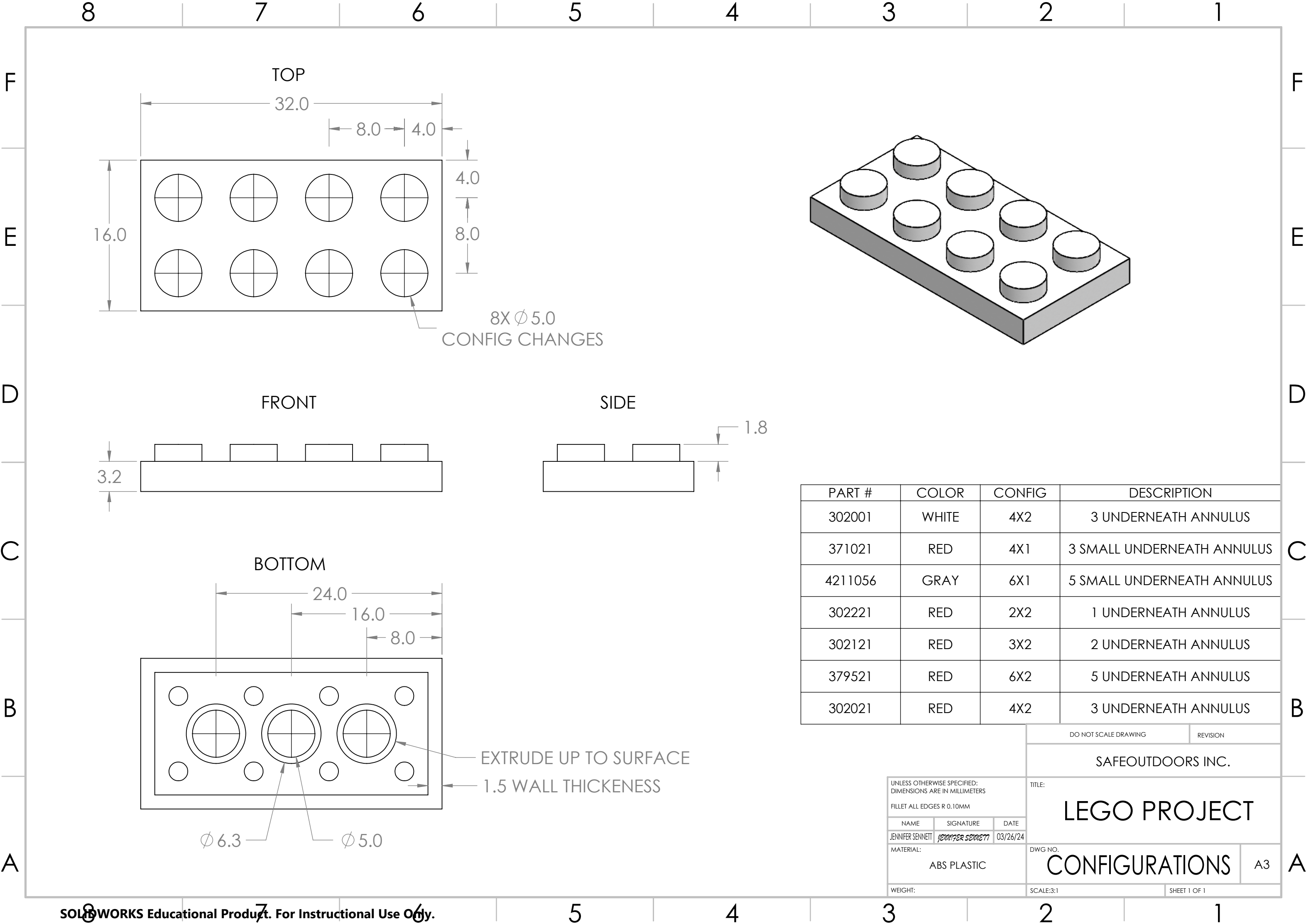
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1.5 WALL THICKNESS



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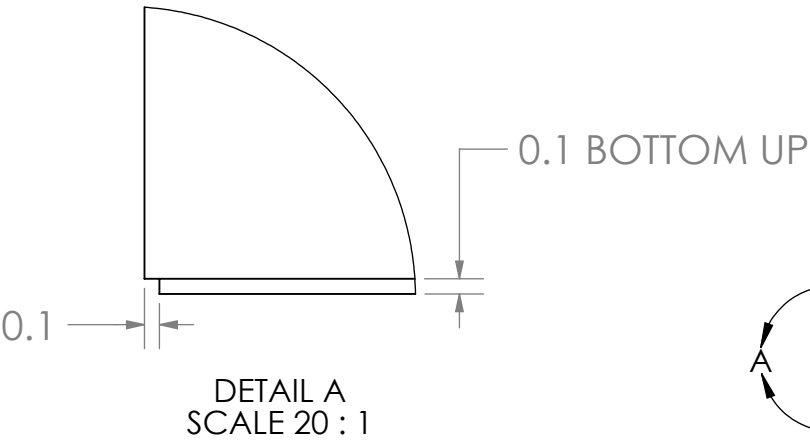
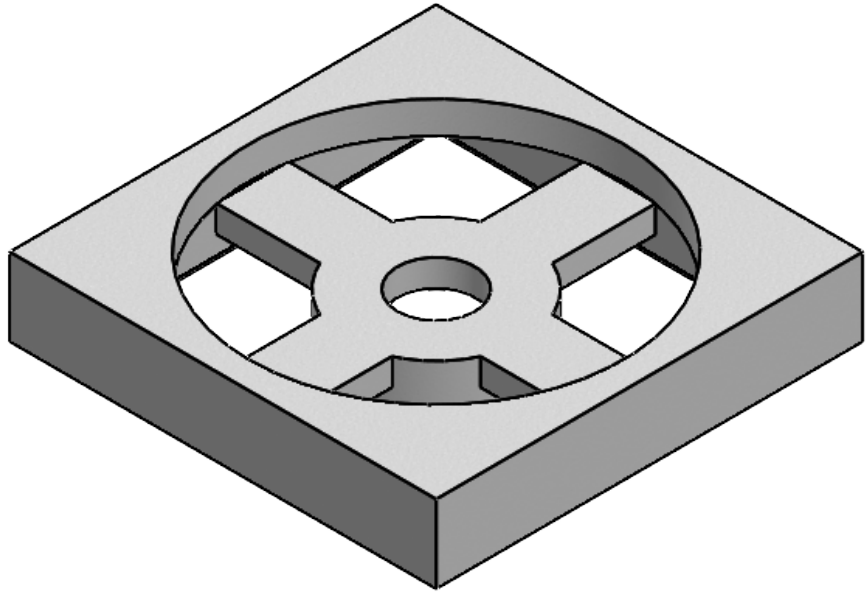
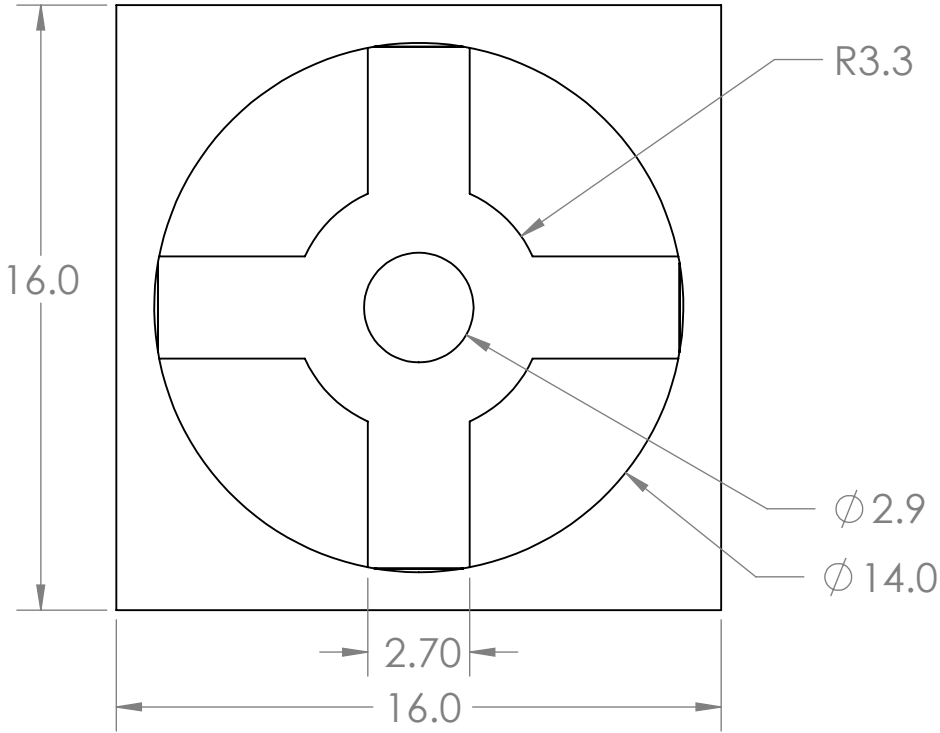
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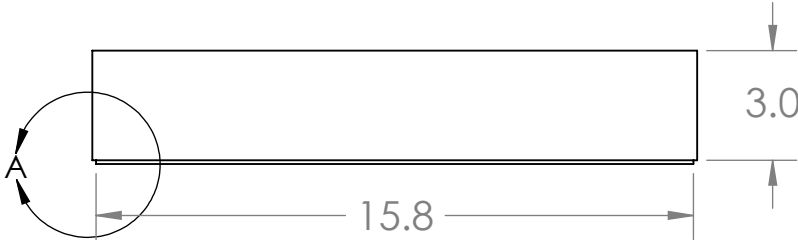
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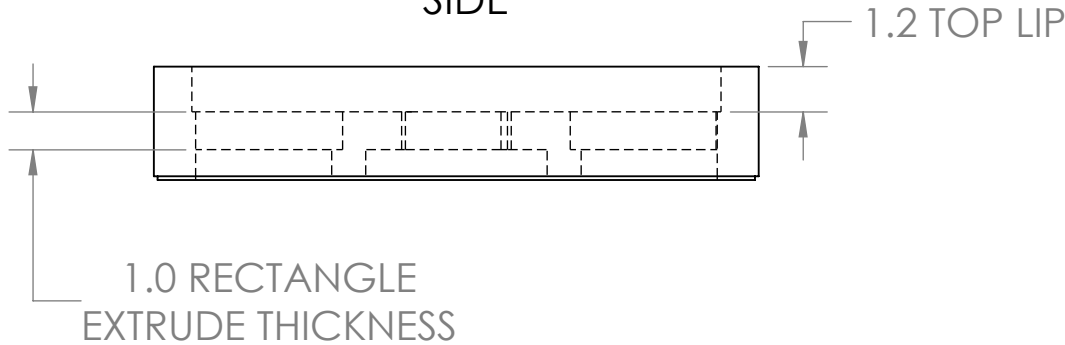
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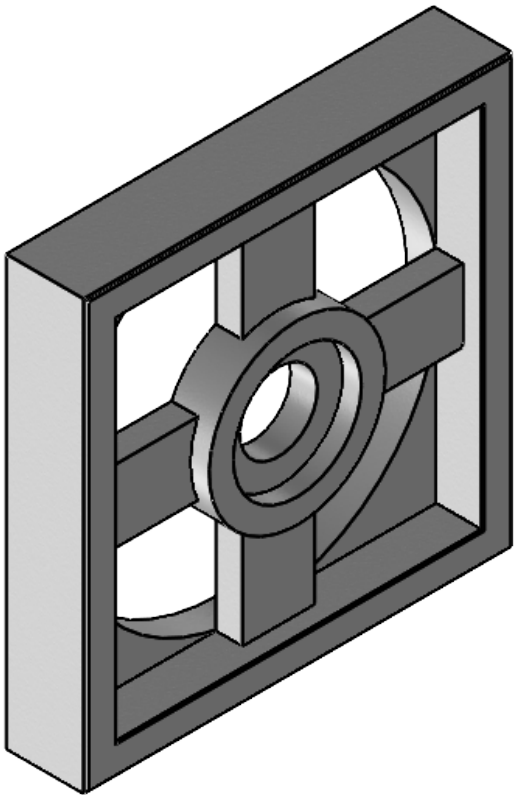
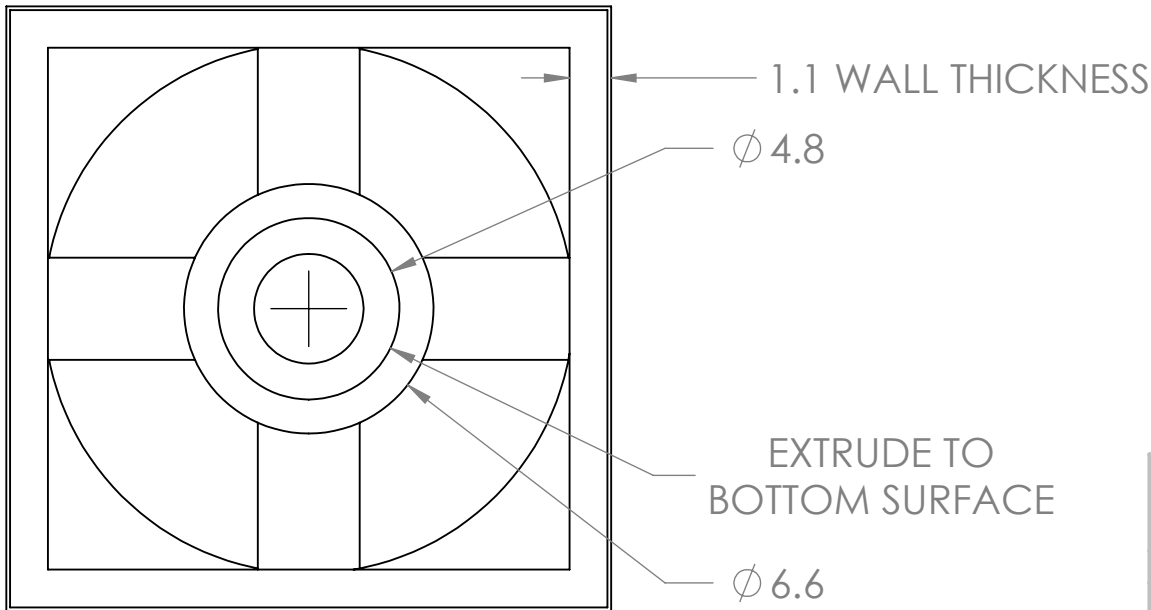
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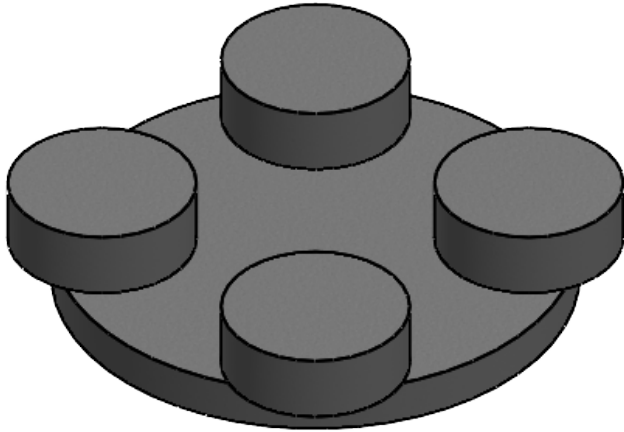
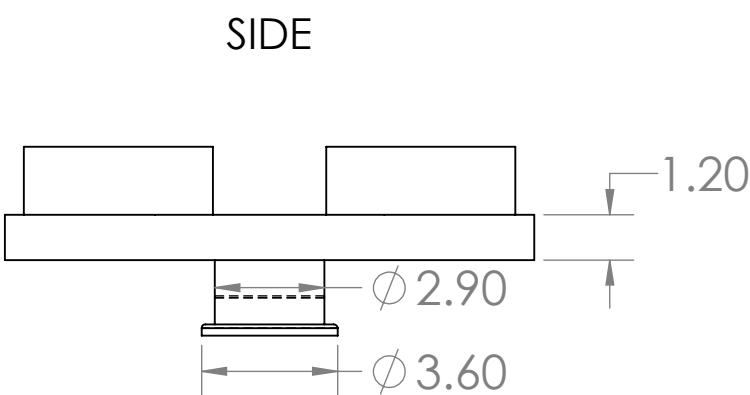
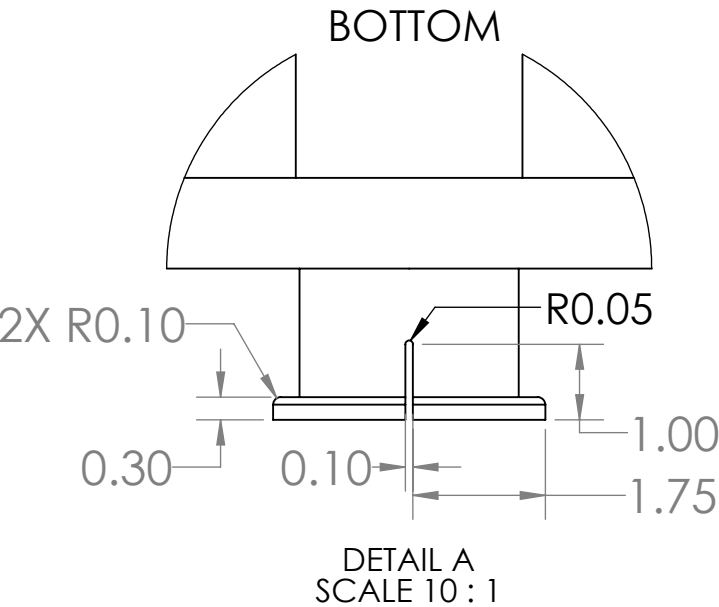
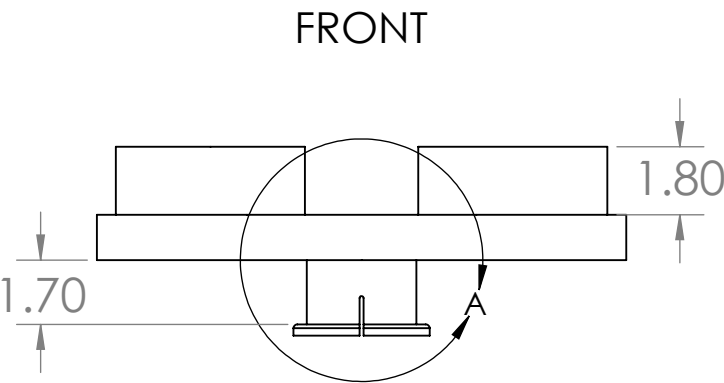
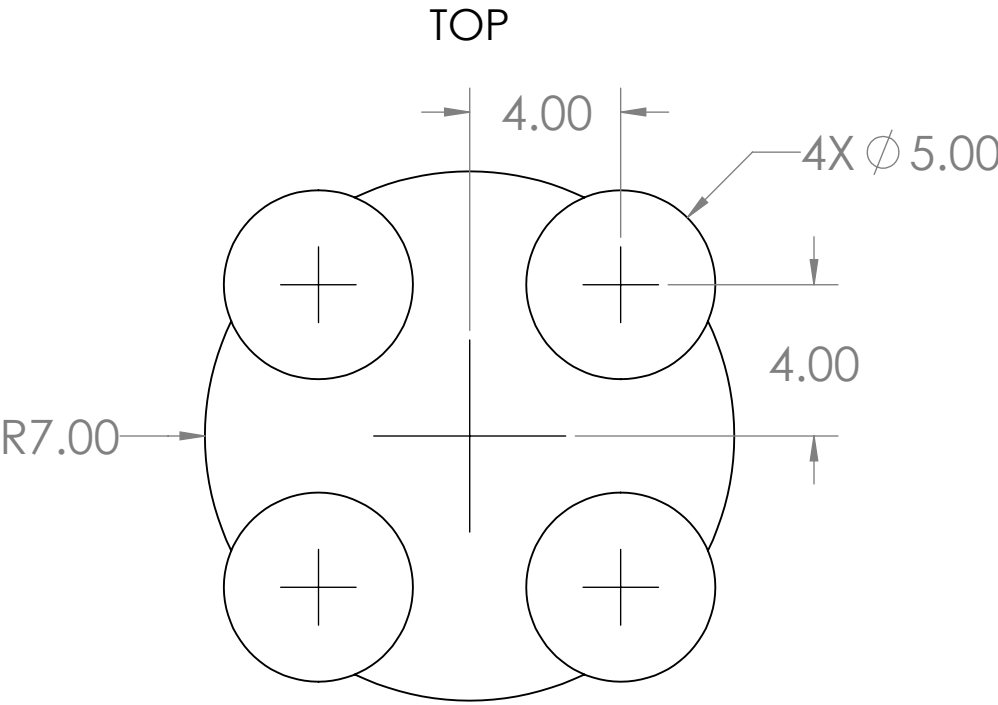
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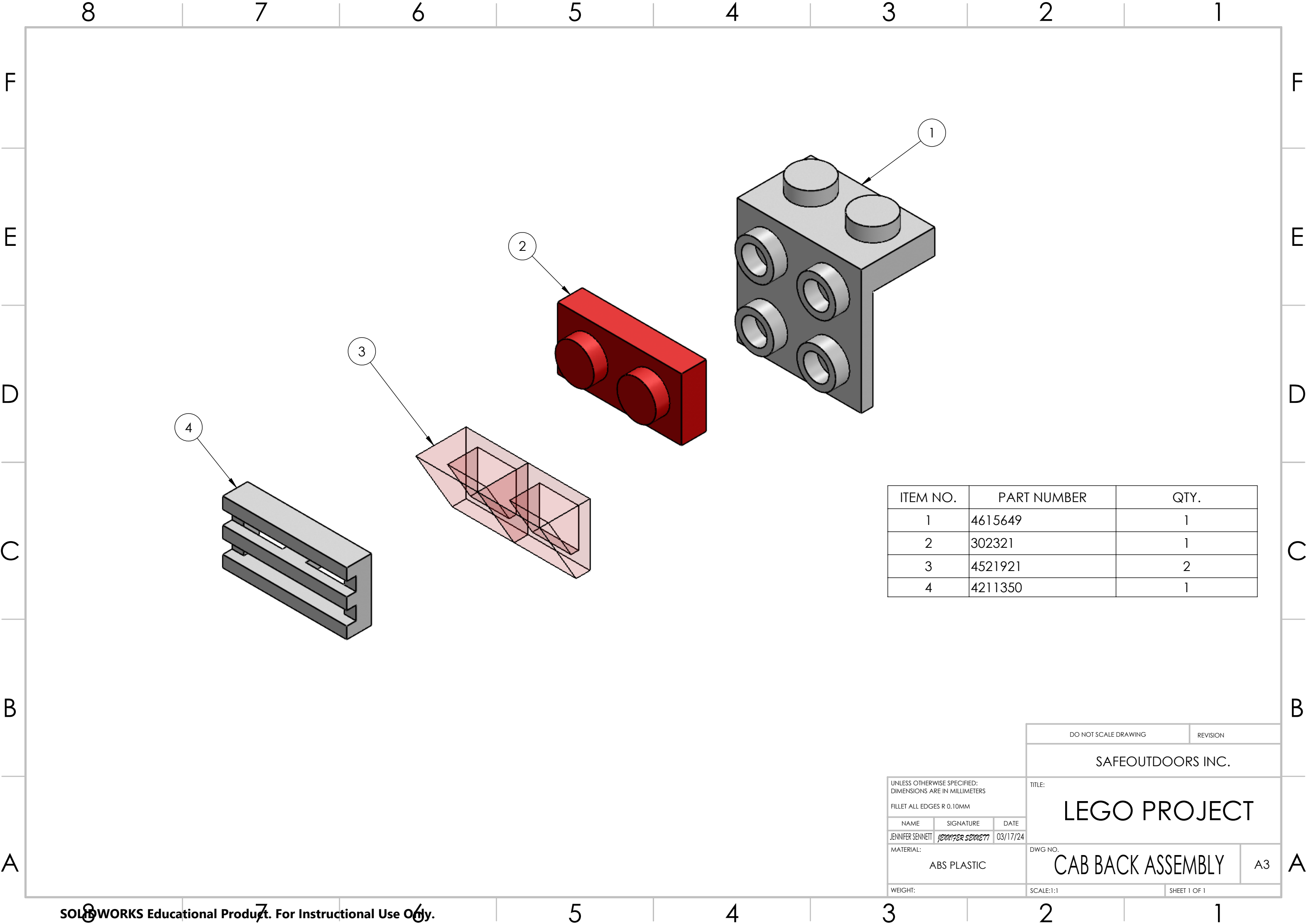
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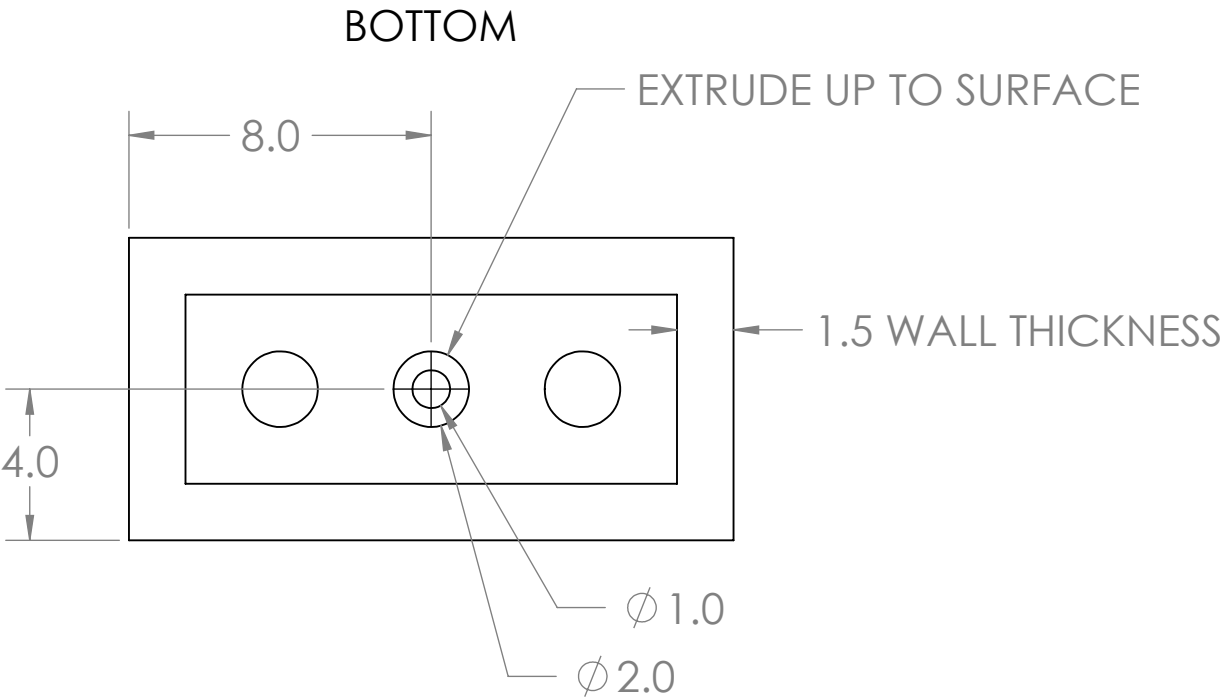
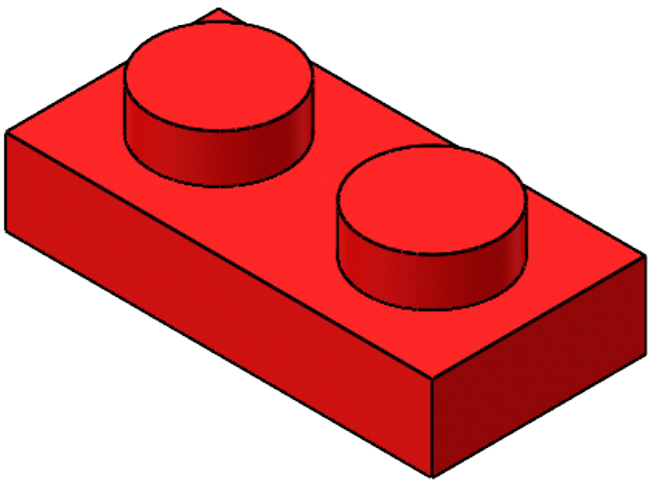
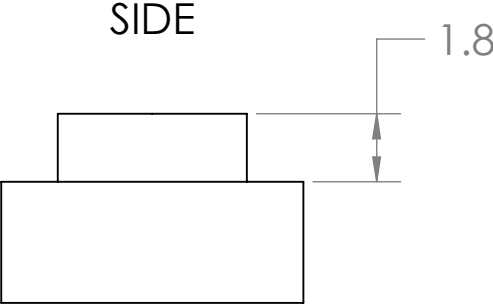
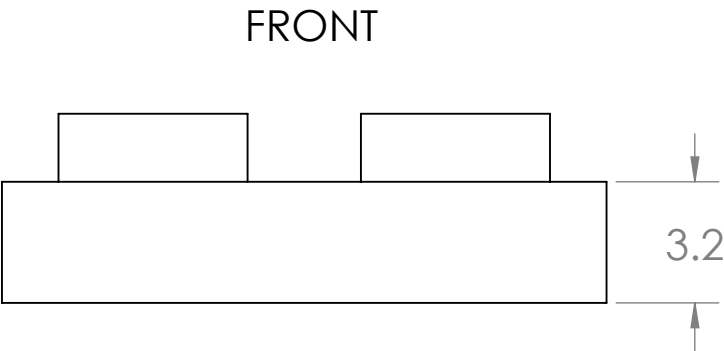
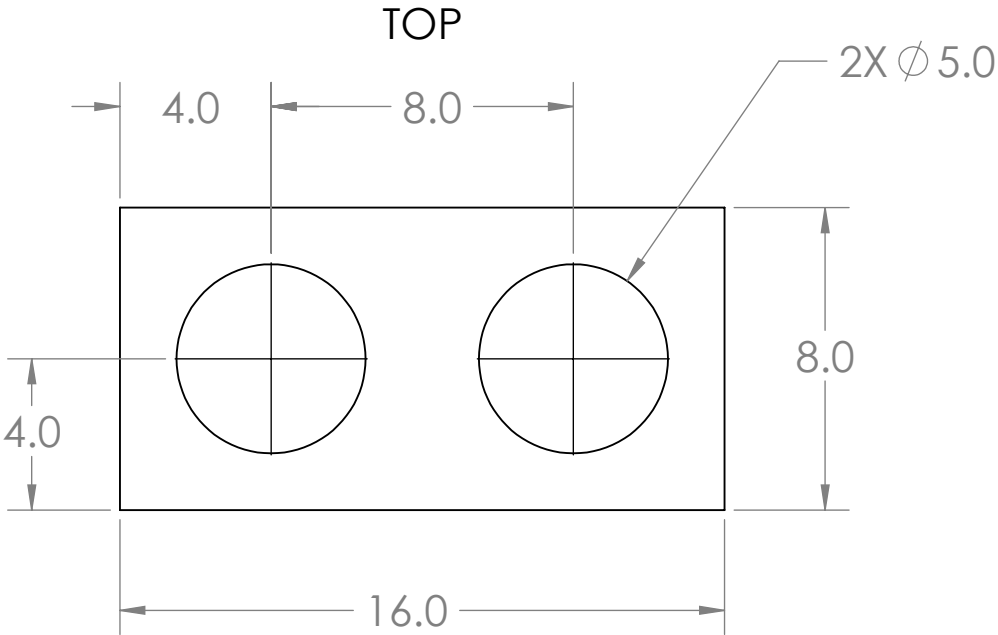
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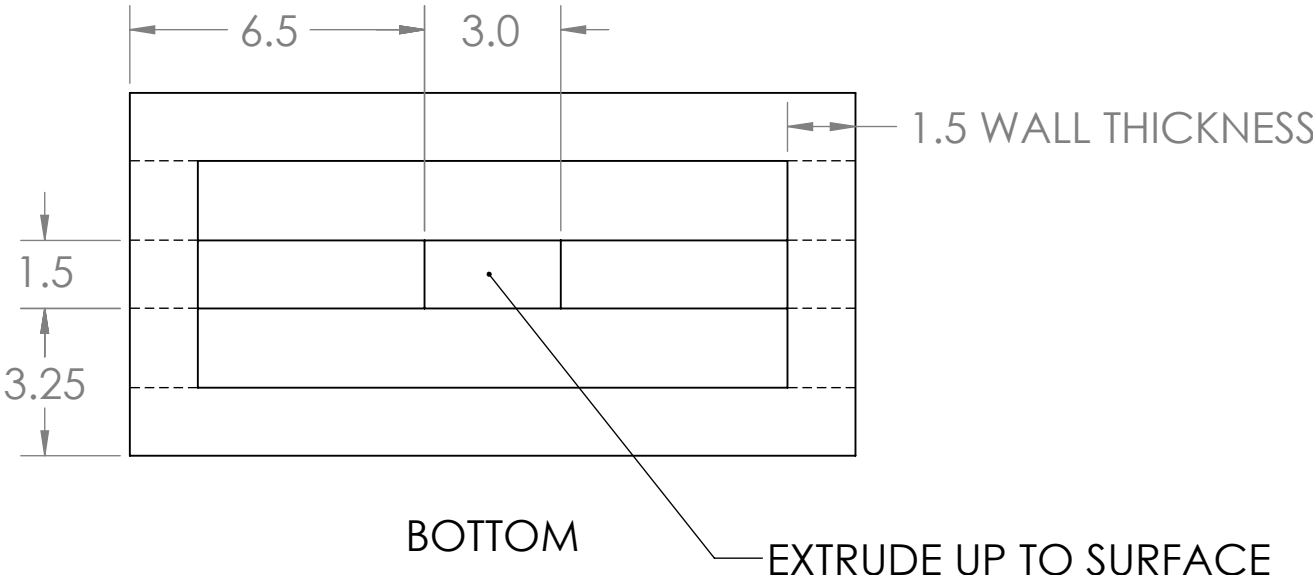
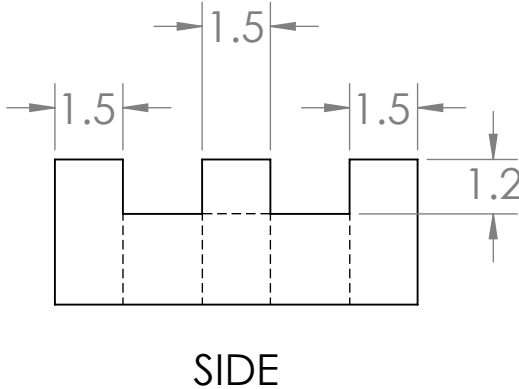
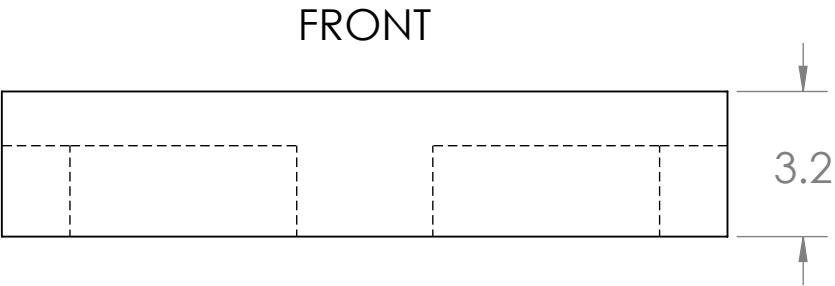
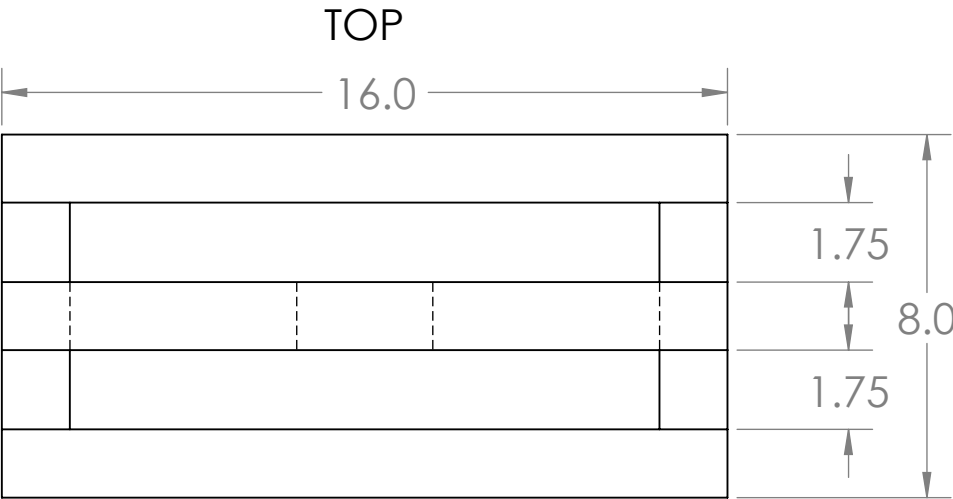
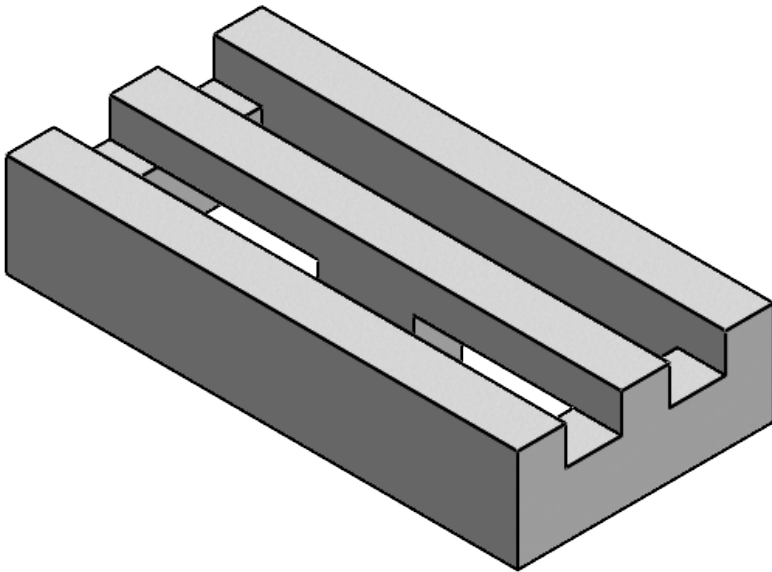
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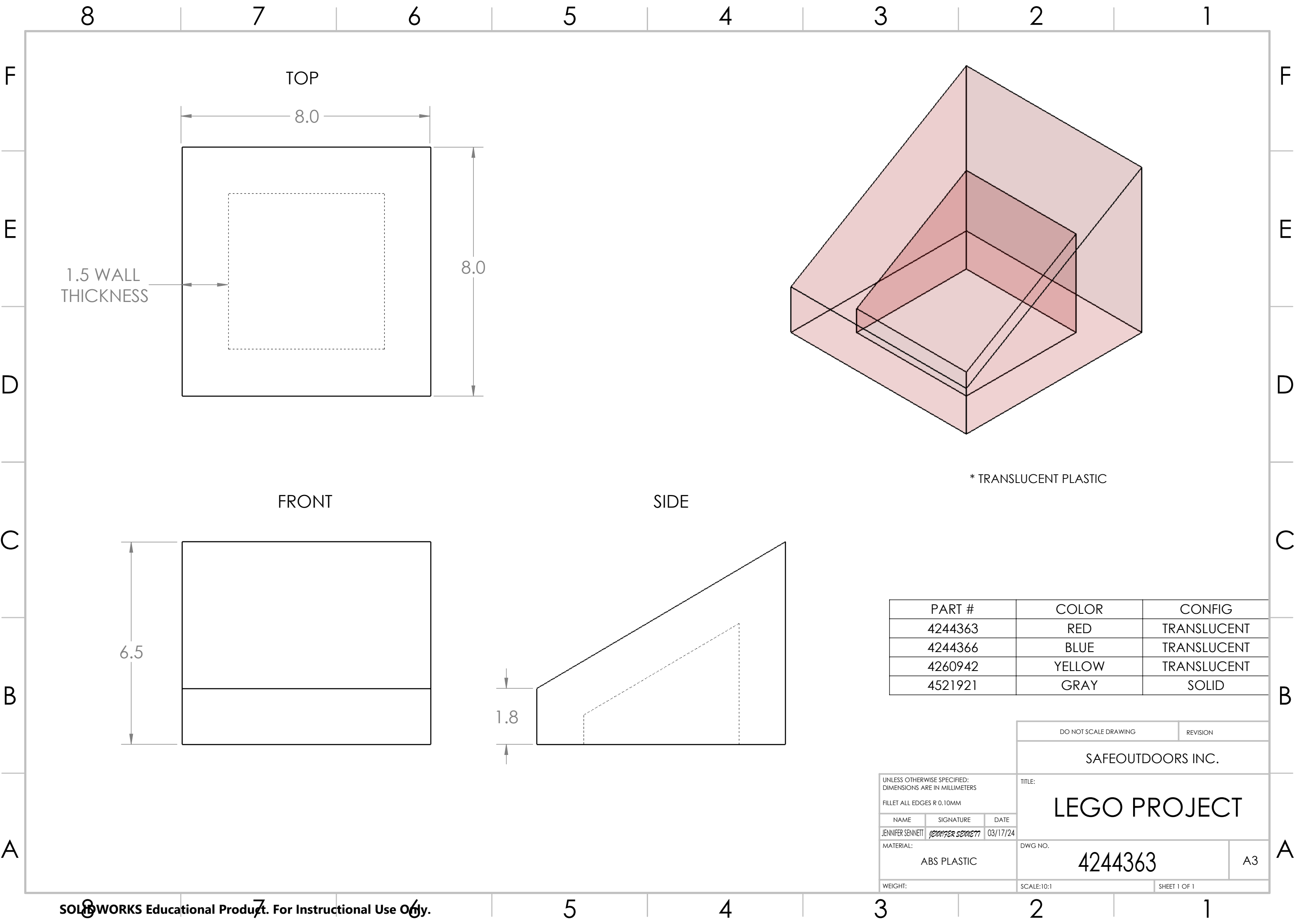
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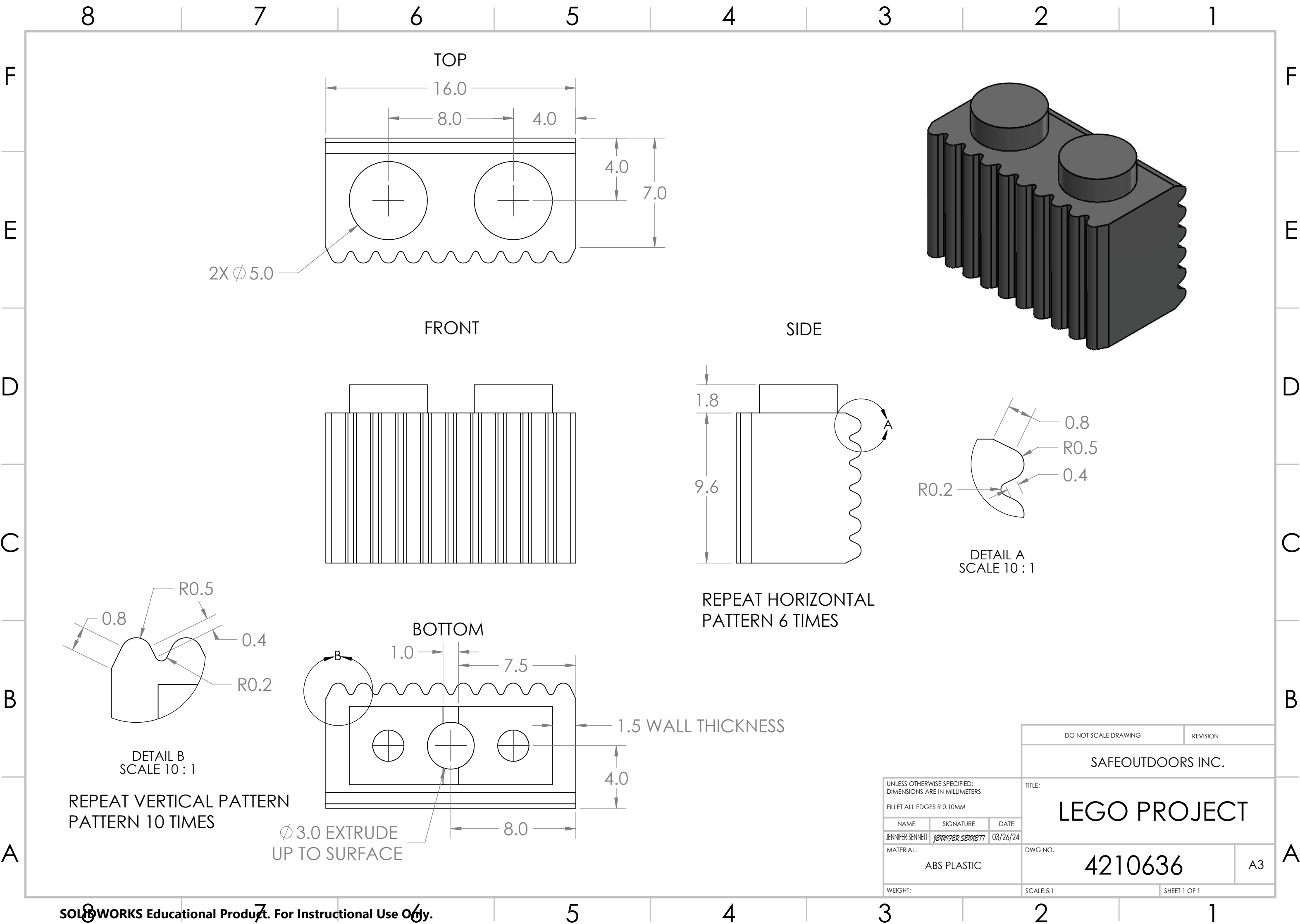
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TITLE:		
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PART #	COLOR	CONFIG
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4244366	BLUE	TRANSLUCENT
4260942	YELLOW	TRANSLUCENT
4521921	GRAY	SOLID

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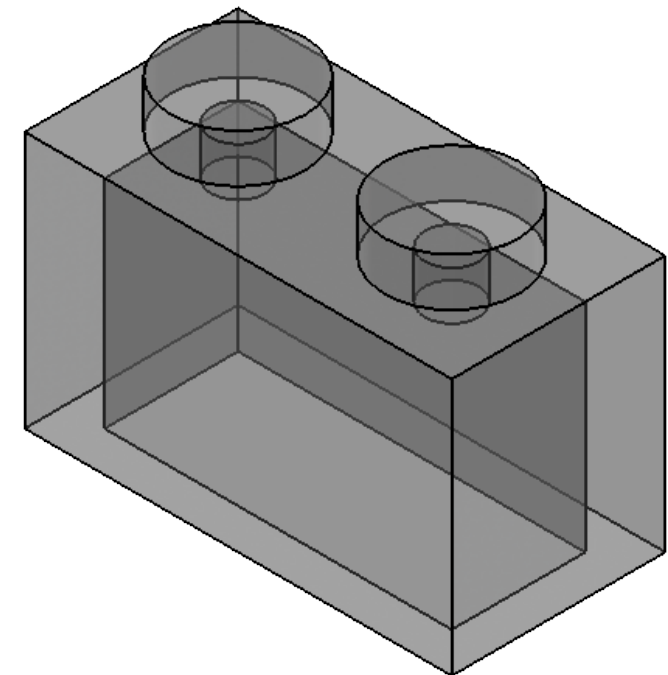
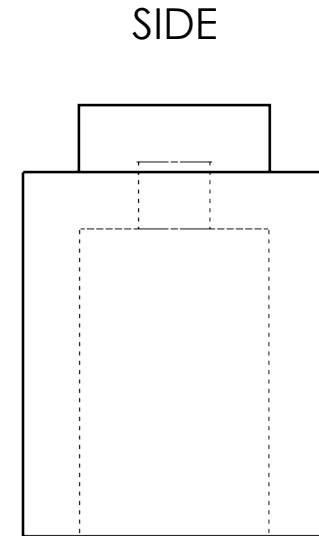
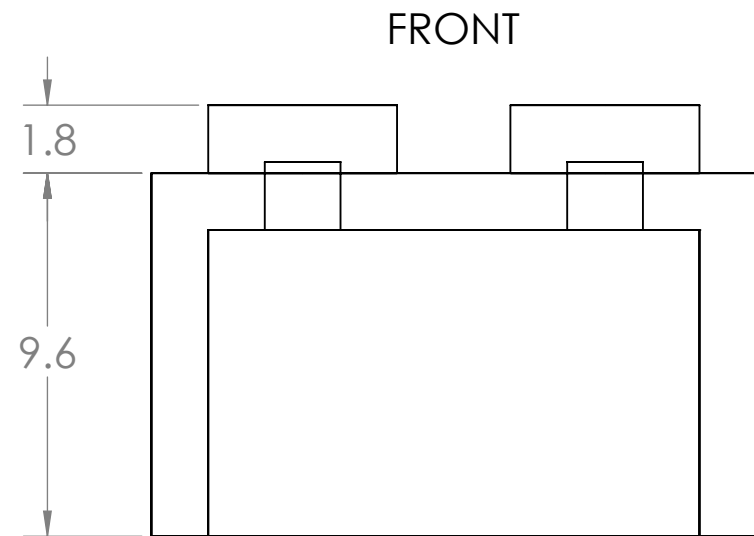
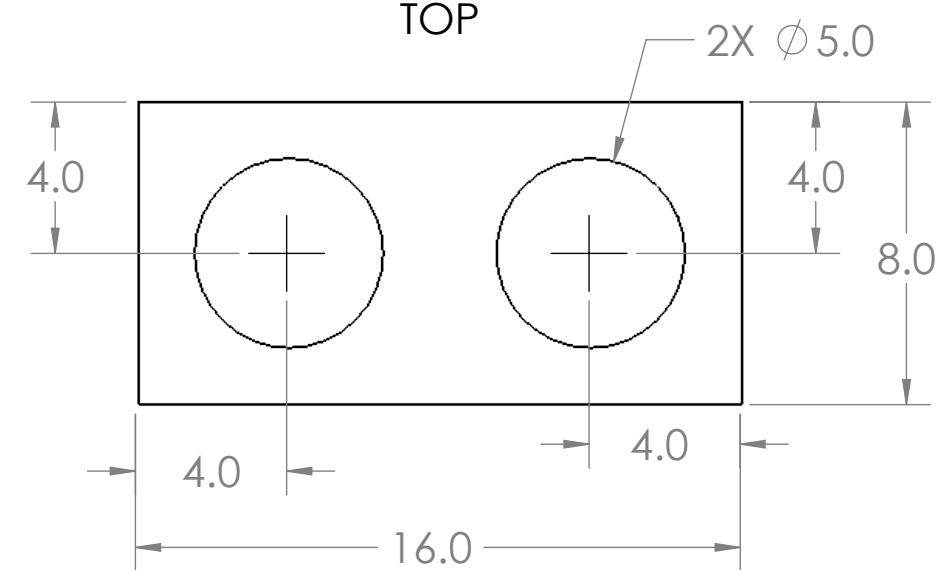
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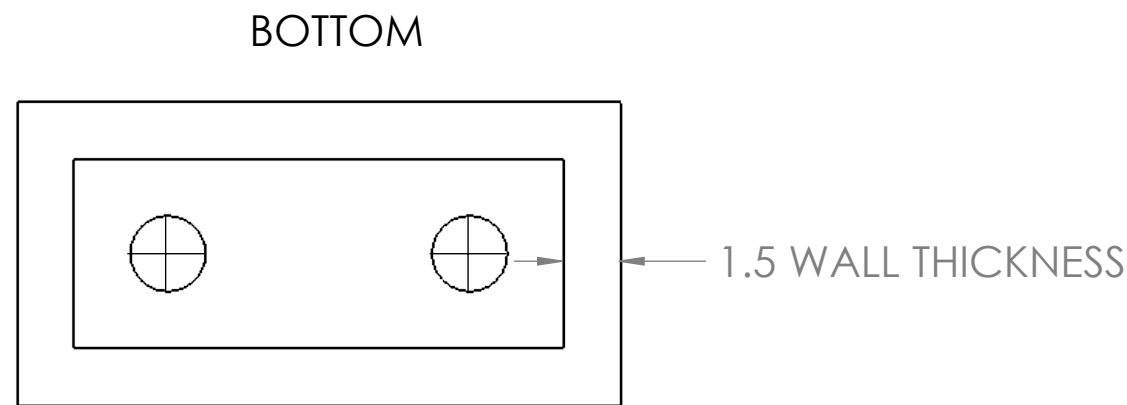
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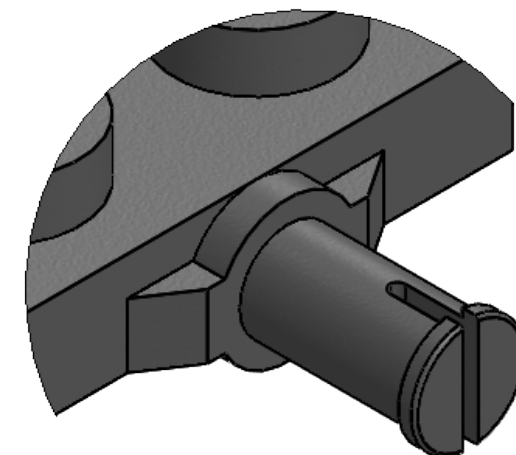
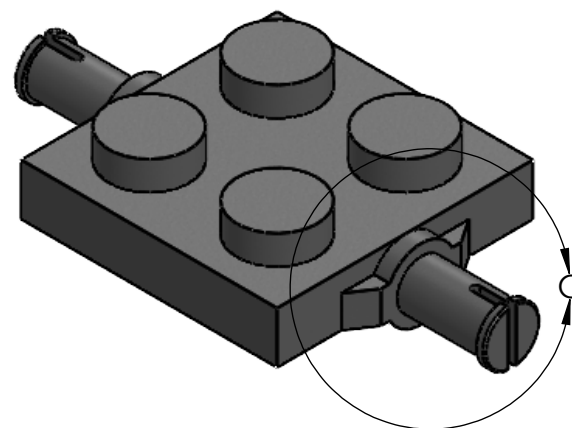
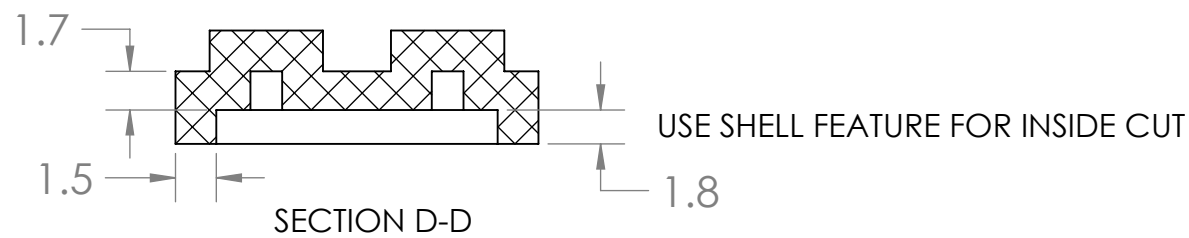
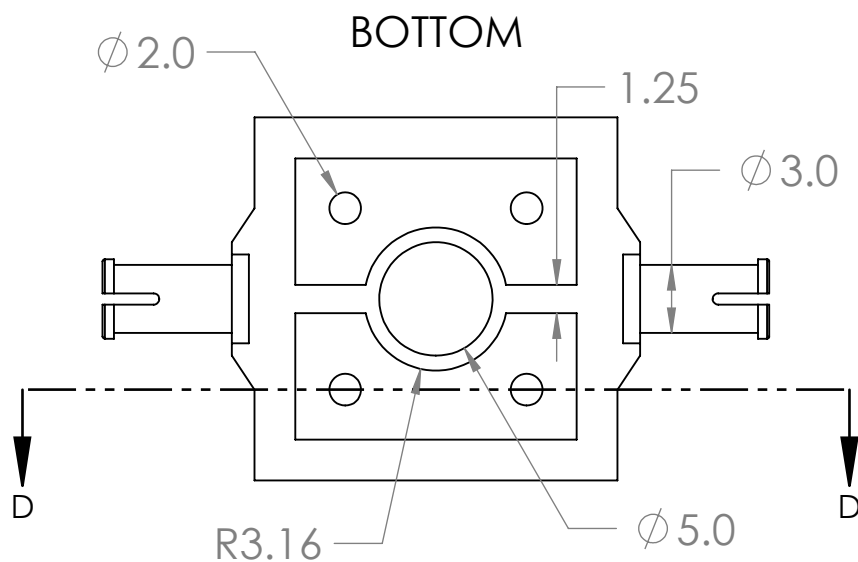
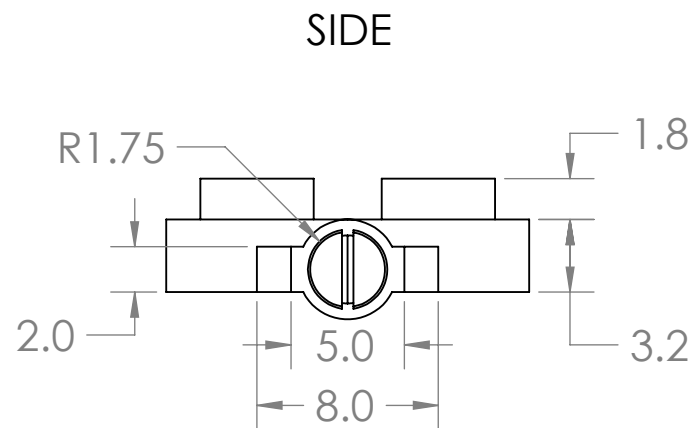
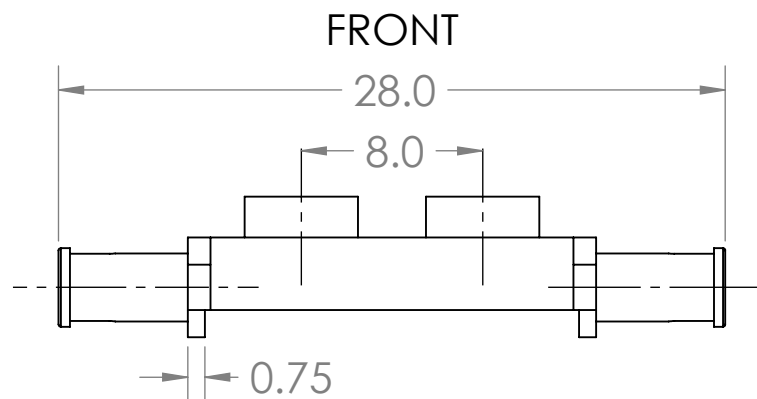
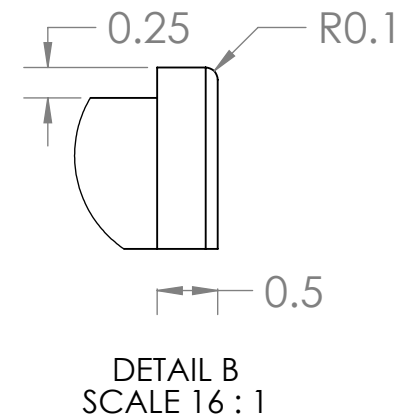
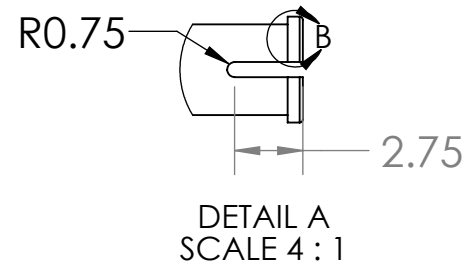
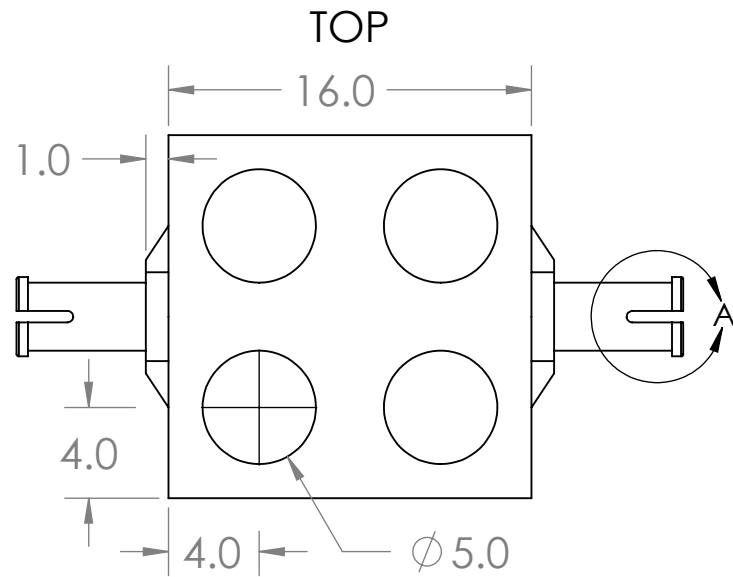


* Translucent Gray



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MATERIAL:		
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WEIGHT:		

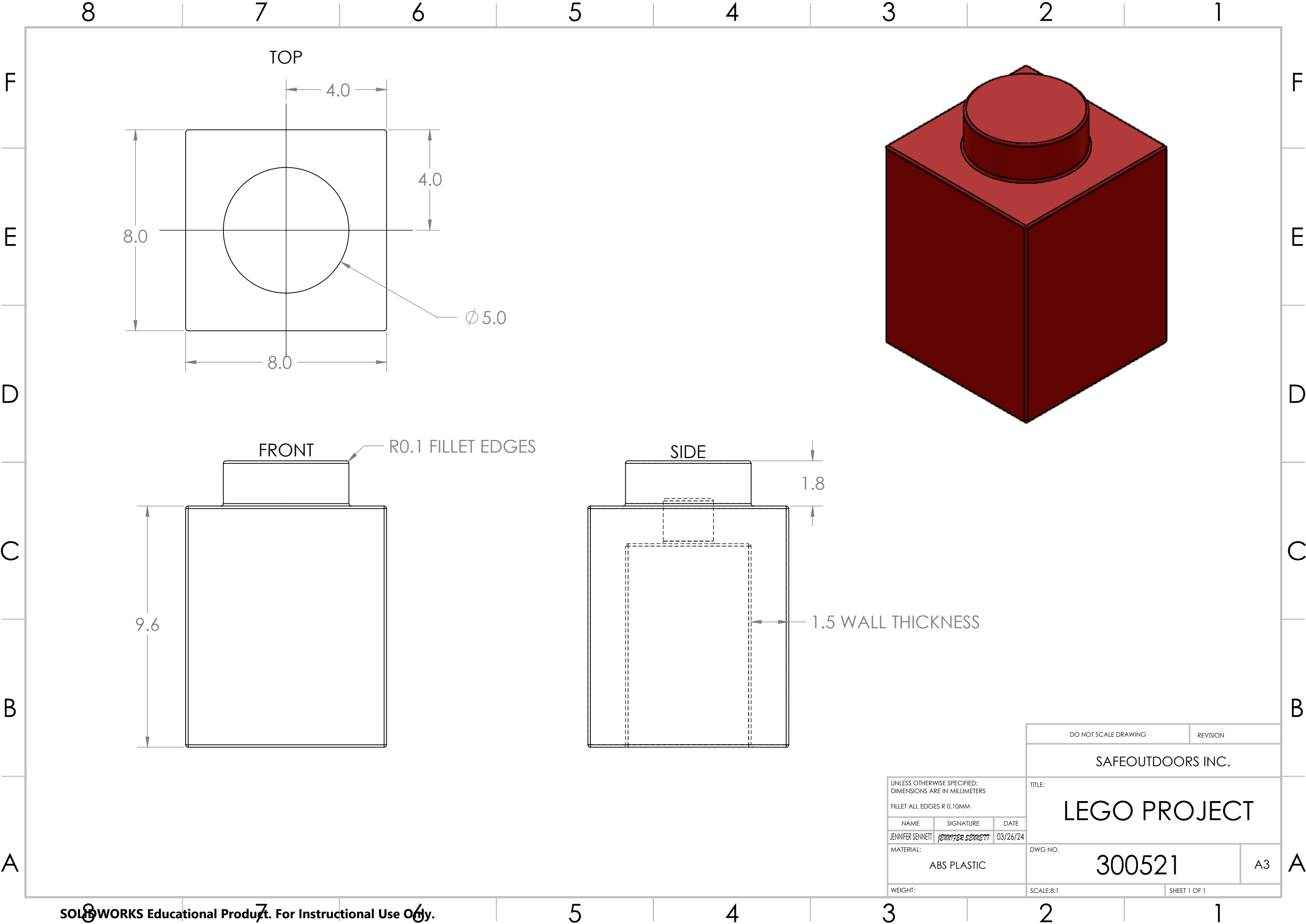
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DETAIL C
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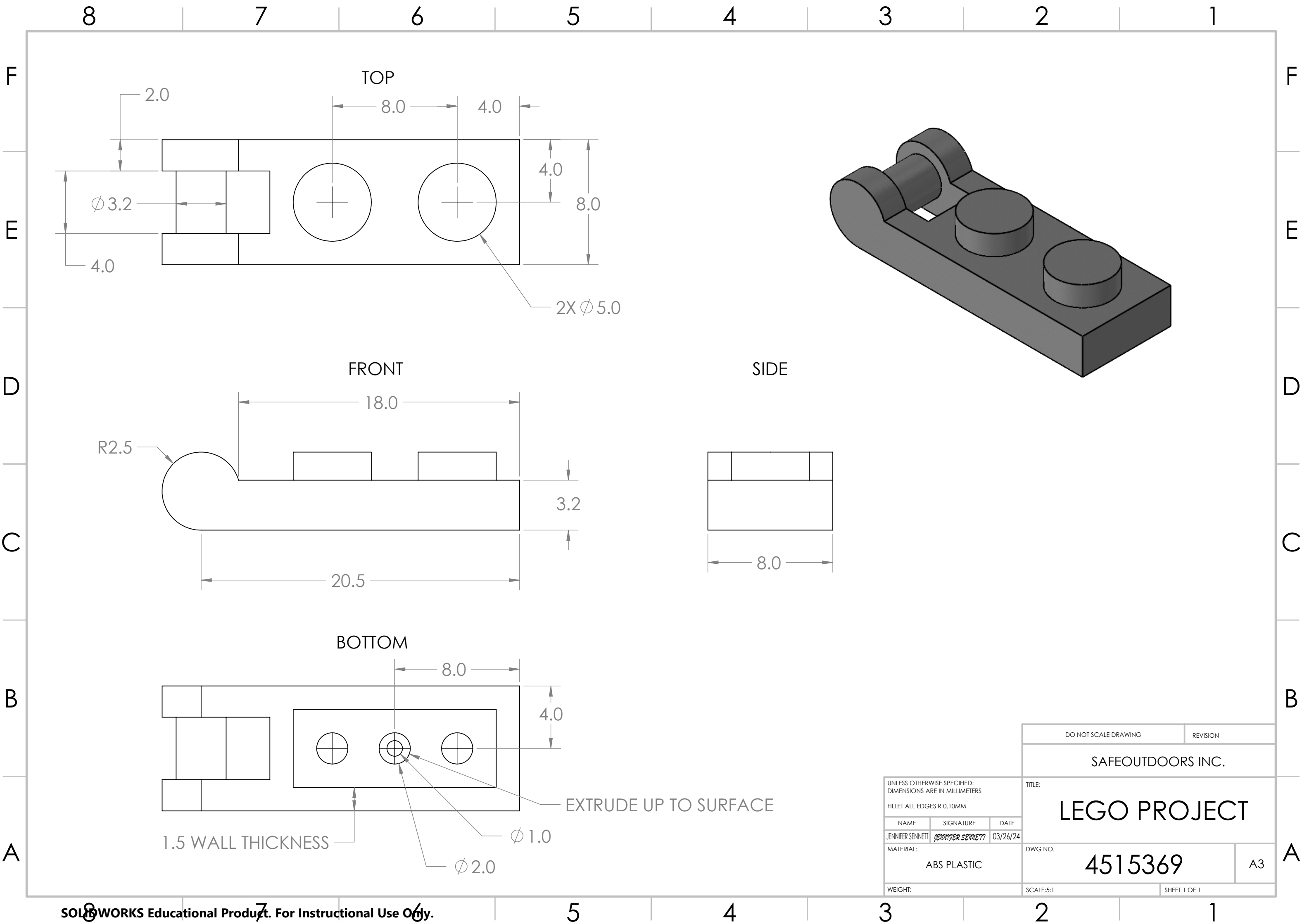
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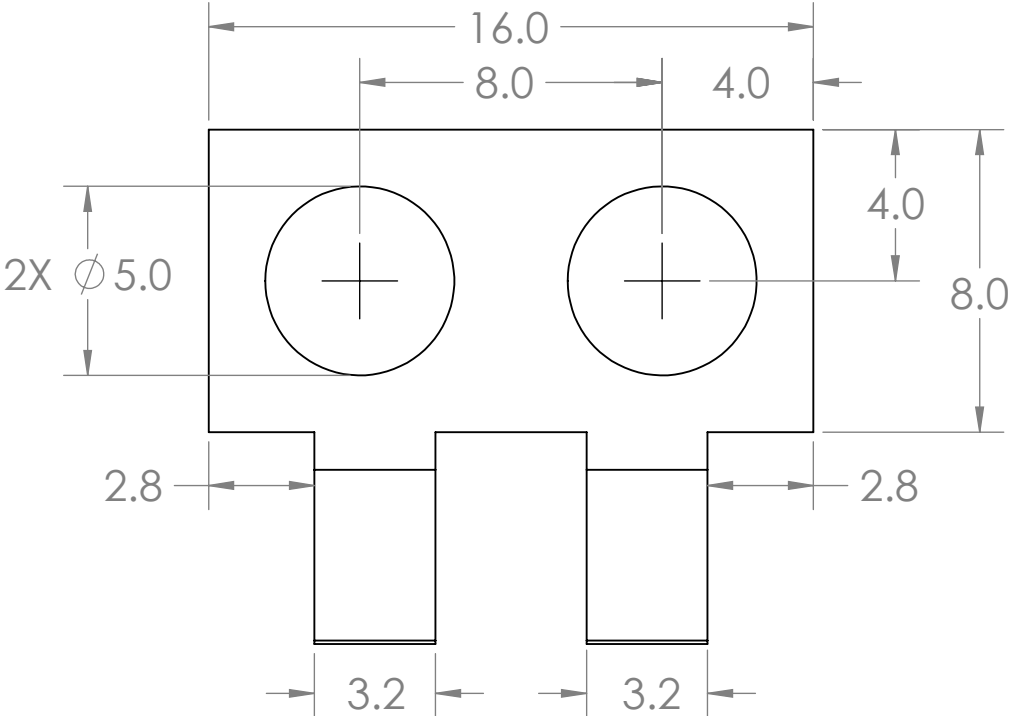
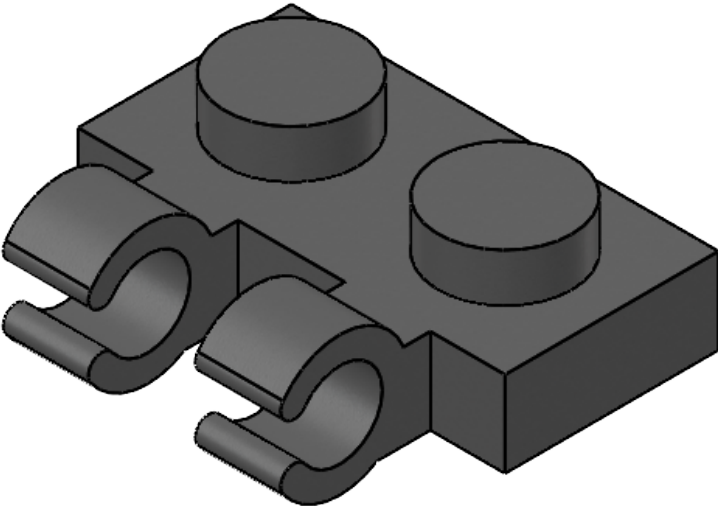
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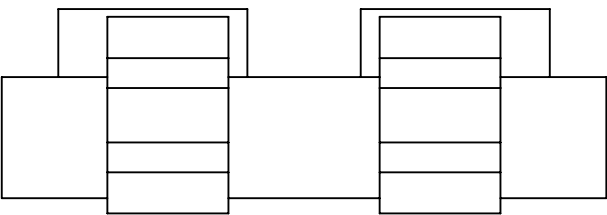
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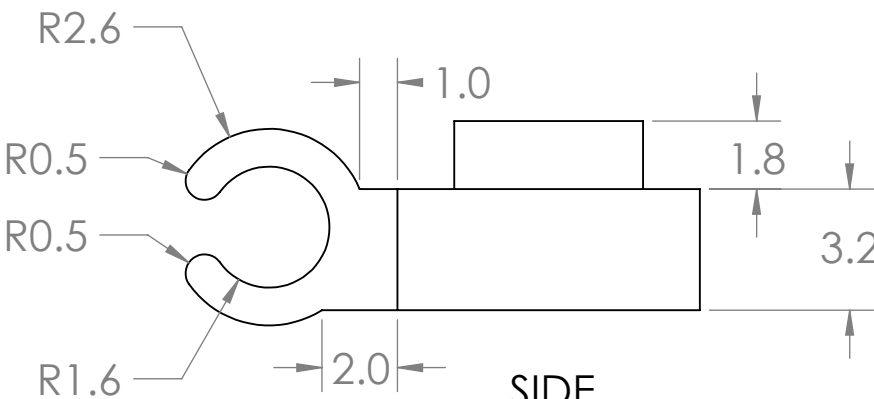
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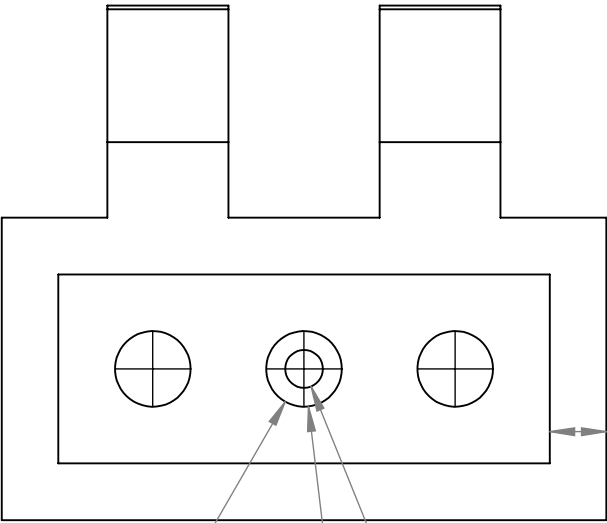
TOP



FRONT



SIDE



BOTTOM

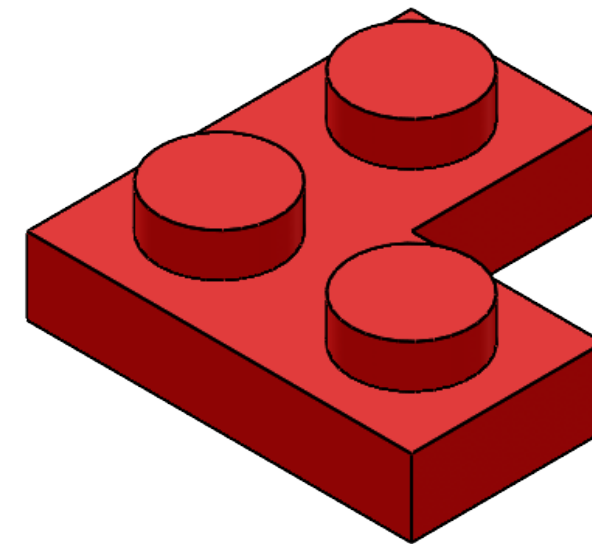
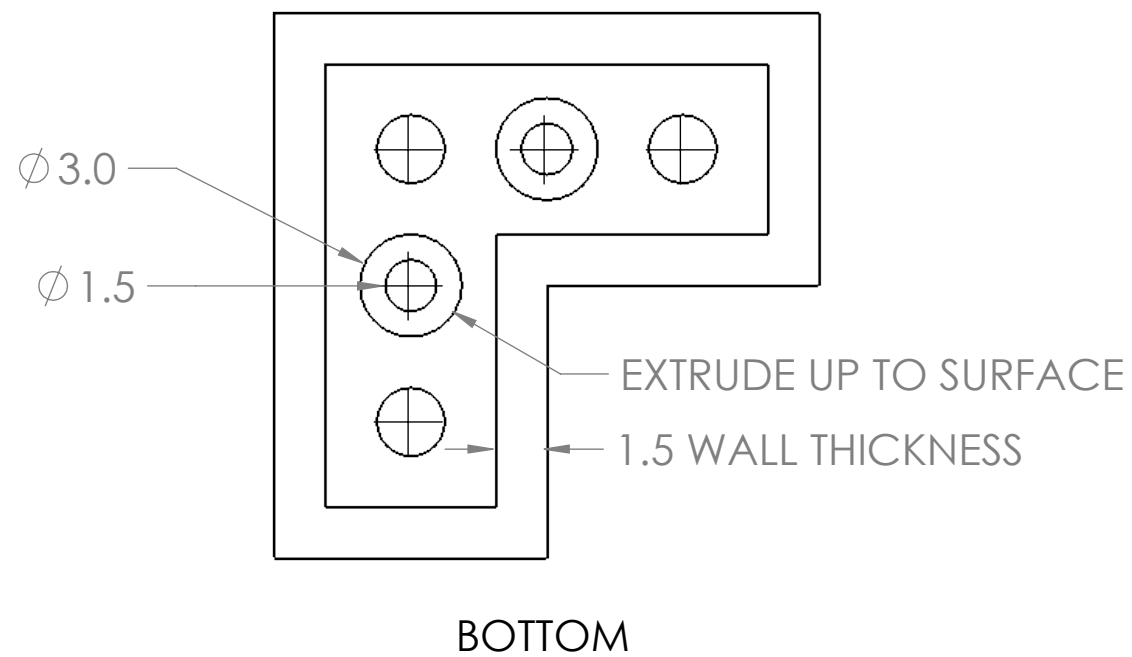
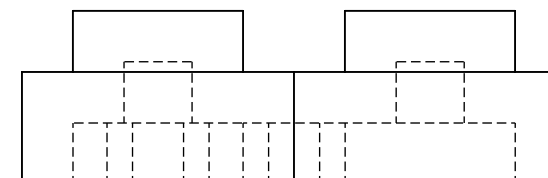
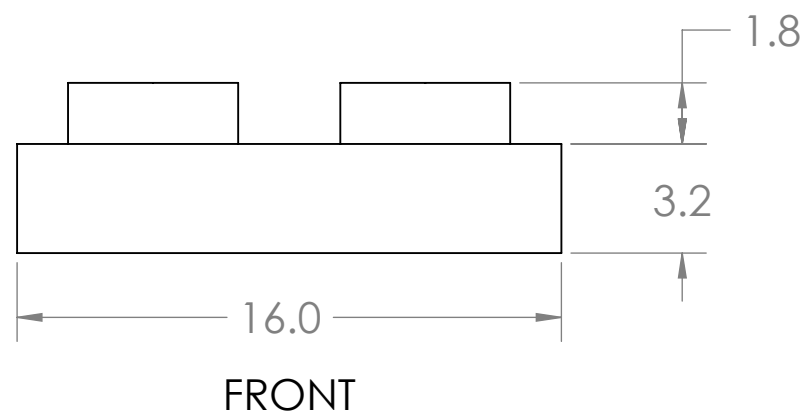
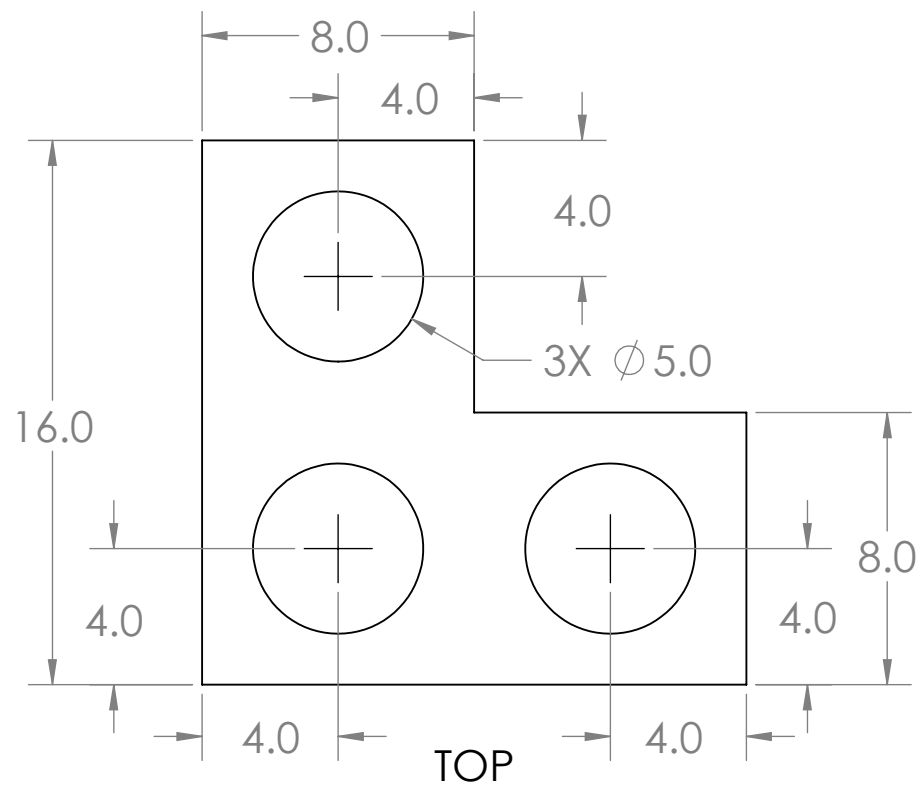
EXTRUDE TO SURFACE

1.5 WALL THICKNESS

Ø 1.0
Ø 2.0

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NAME	SIGNATURE	DATE
JENNIFER SENNETT	JENNIFER SENNETT	03/26/24
MATERIAL:		
ABS PLASTIC		
WEIGHT:		

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SAFEOUTDOORS INC.		
TITLE: LEGO PROJECT		
DWG NO. 4556157		A3
SCALE:5:1		SHEET 1 OF 1



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DIMENSIONS ARE IN MILLIMETERS

FILLET ALL EDGES R 0.10MM

NAME	SIGNATURE	DATE
JENNIFER SENNETT	<i>JENNIFER SENNETT</i>	03/17/24

MATERIAL:

ABS PLASTIC

WEIGHT:

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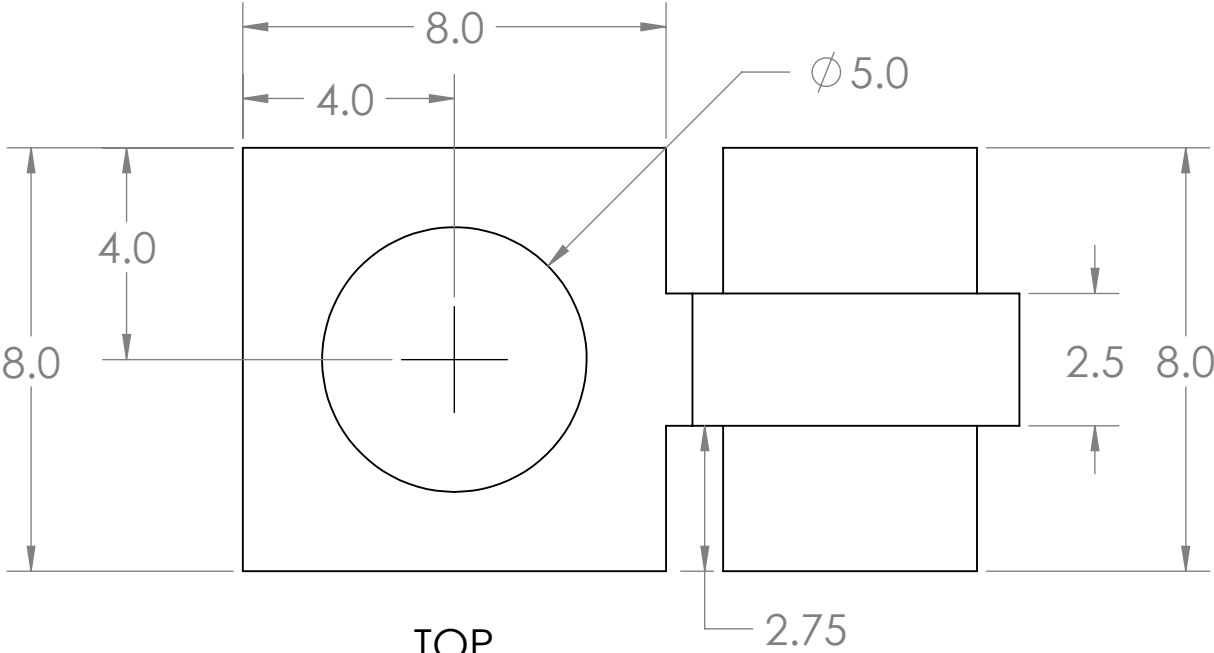
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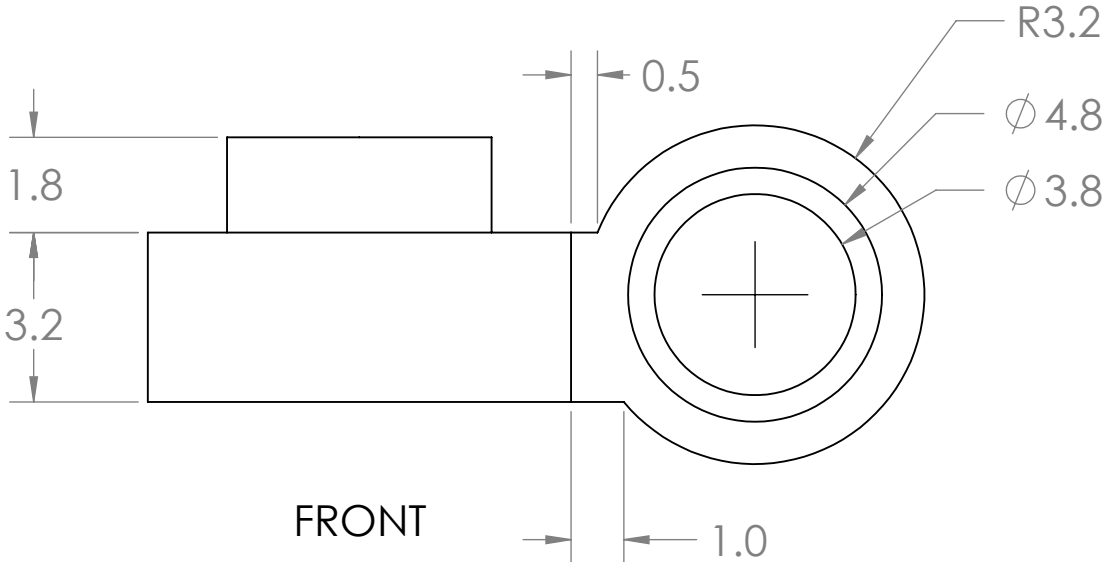
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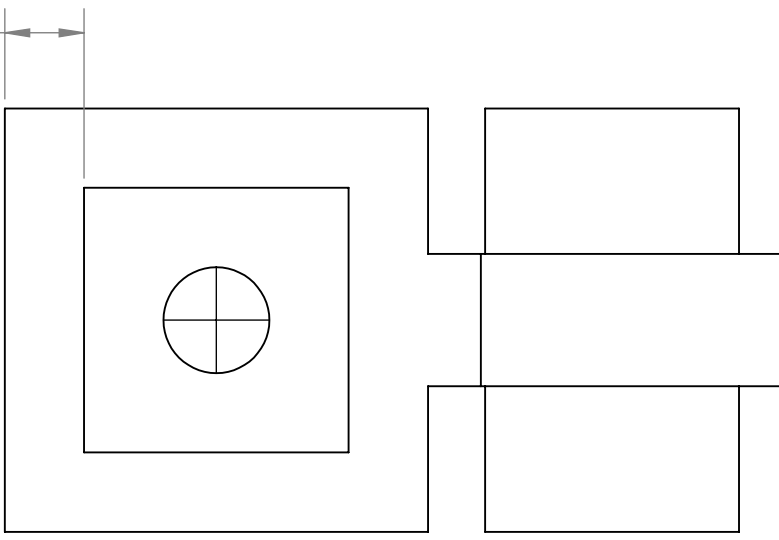


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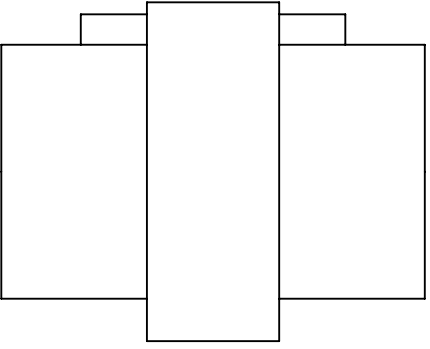


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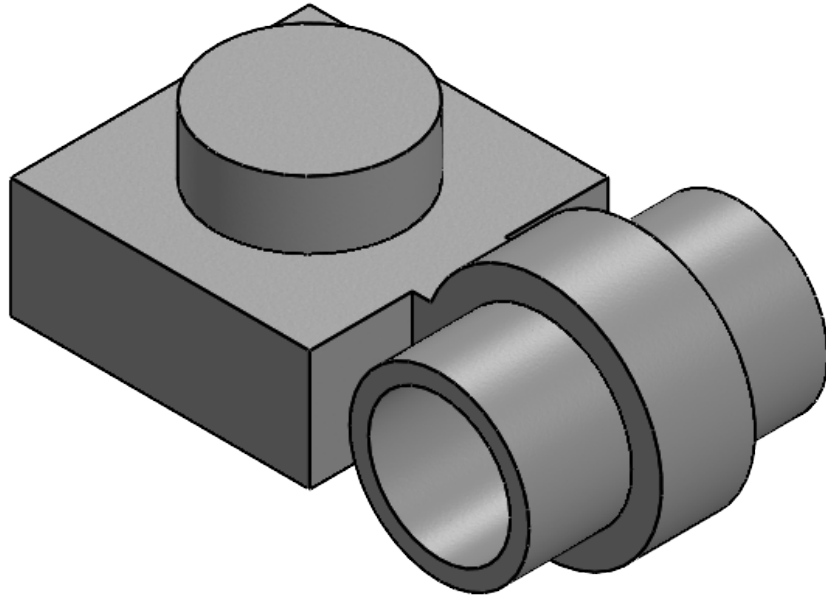
1.5 WALL THICKNESS



BOTTOM



SIDE



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS		
FILLET ALL EDGES R 0.10MM		
NAME	SIGNATURE	DATE
JENNIFER SENNETT	JENNIFER SENNETT	03/26/24
MATERIAL:		
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SAFEOUTDOORS INC.		
TITLE:		
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4632575		
SCALE:7:1		SHEET 1 OF 1

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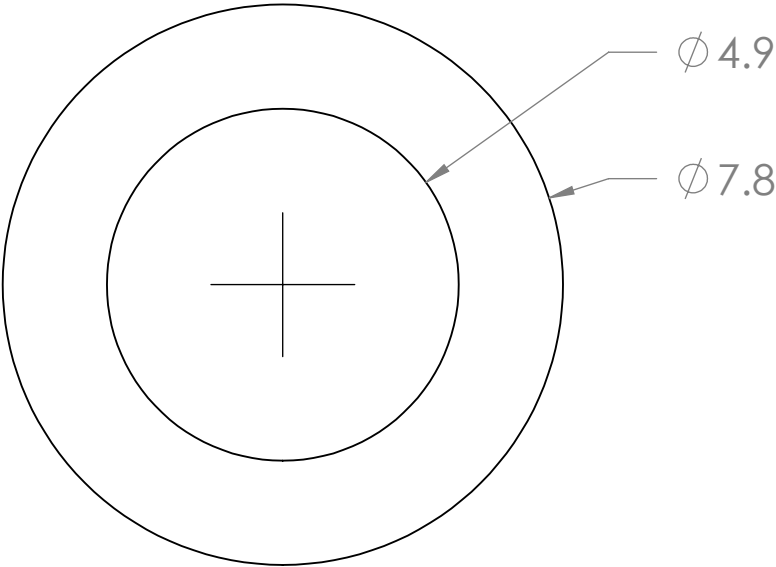
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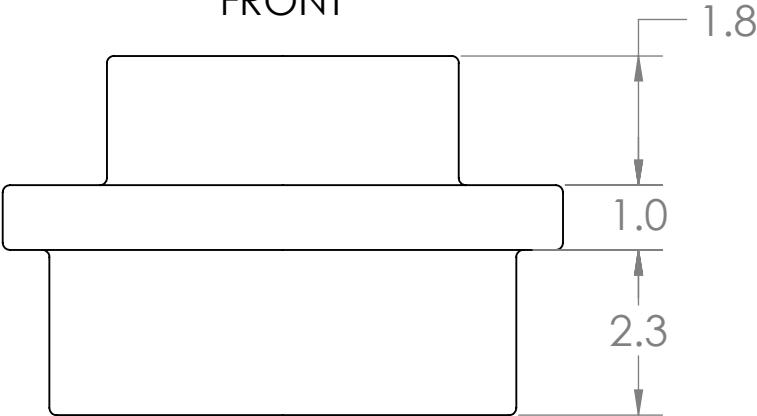
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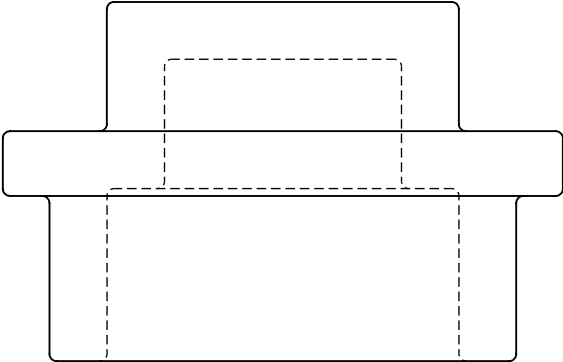
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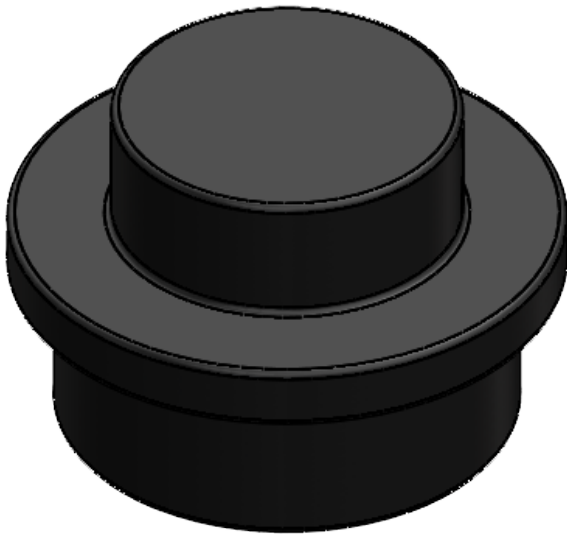
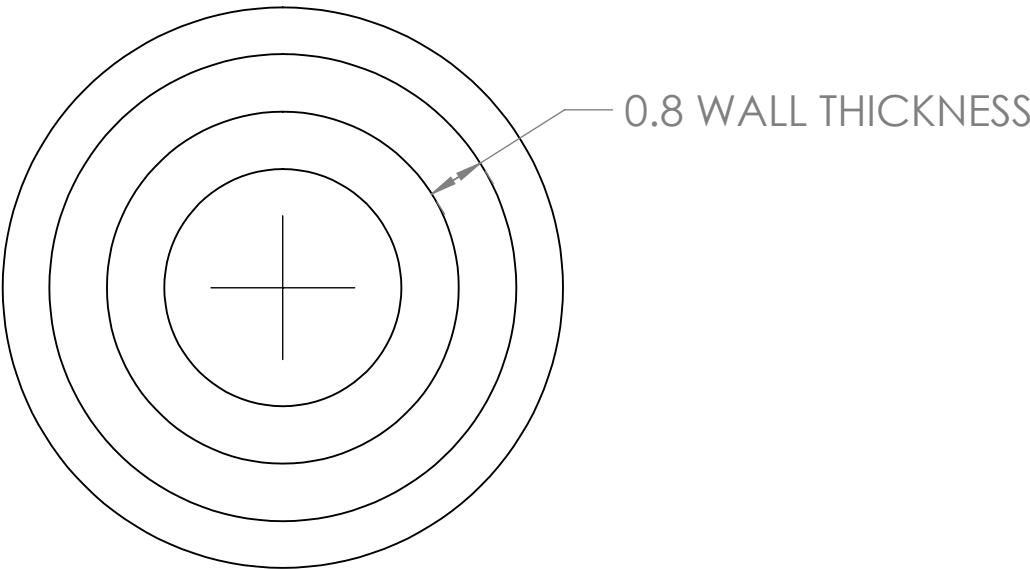
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UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS		
FILLET ALL EDGES R 0.10MM		
NAME	SIGNATURE	DATE
JENNIFER SENNETT	JENNIFER SENNETT	03/17/24
MATERIAL:		
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SAFEOUTDOORS INC.		
TITLE:		
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DWG NO.		A3
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